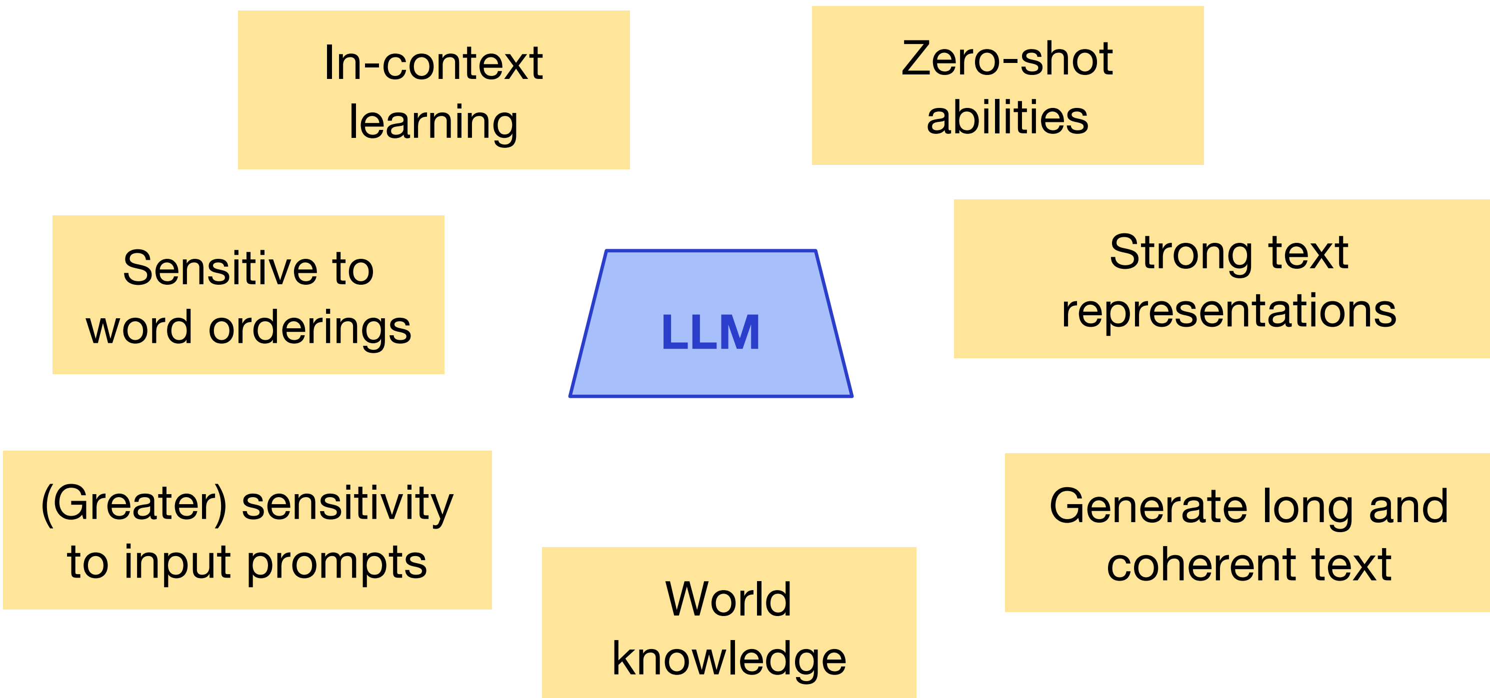
Multimodal Models

Russ Salakhutdinov and Jing Yu Koh

Machine Learning Department
Carnegie Mellon University

Large Language Models



What is Missing?

- Image + text inputs → text outputs
  - Much more general than text-only LMs
- **Early on (2014 - 2020):** Finetune on paired image-text datasets for specific tasks
- **More recently**: Map a strong vision model to an LLM
  - Use prompting rather than finetuning
  - Leverage the abilities of pretrained LLMs

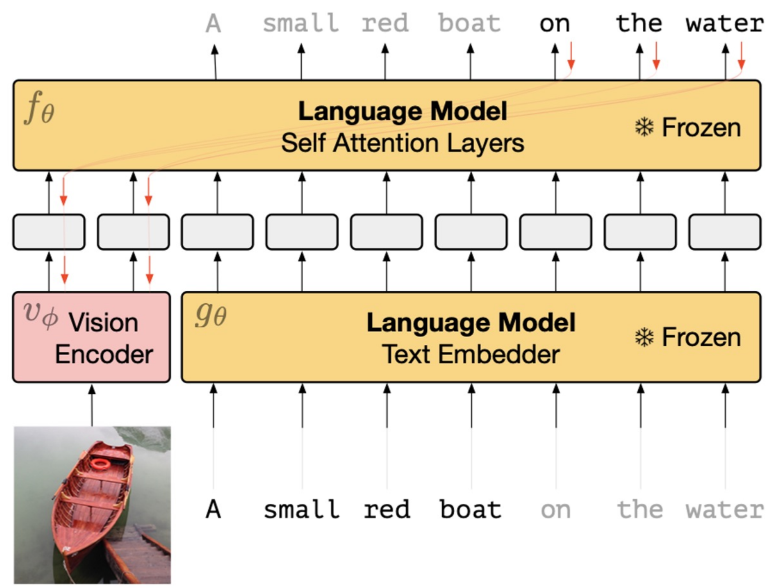


Image to Text Tasks

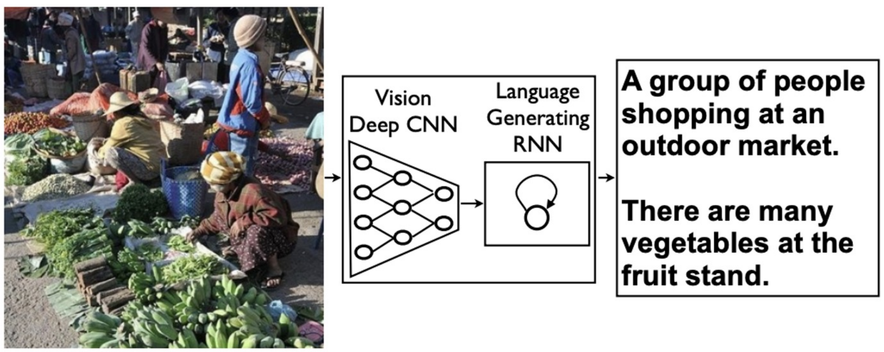
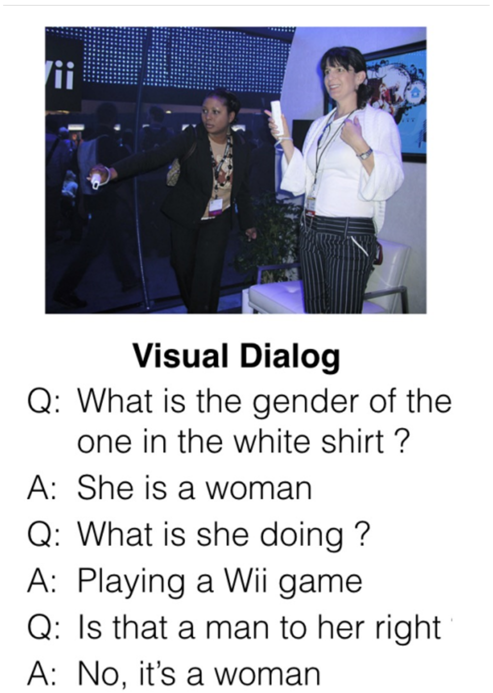


Image Captioning

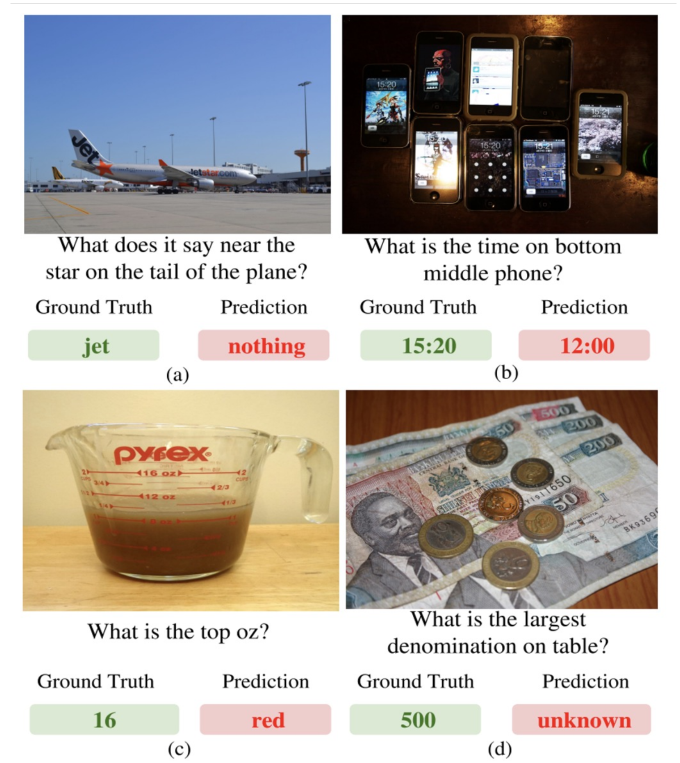
VQA: Visual Question Answering

Show and Tell: A Neural Image Caption Generator ([Vinyals et al., 2014])
VQA: Visual Question Answering ([Agrawal et al., 2016])

Image to Text Tasks



Visual Dialogue

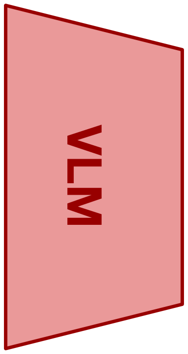


TextVQA

Visual Dialog (Das et al., 2017)
Towards VQA Models That Can Read (Singh et al., 2019)

Image to Text Tasks Image Identification

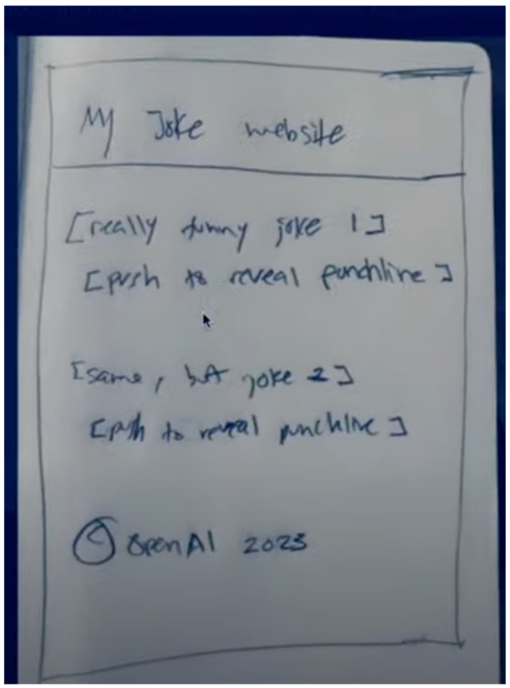
Q: Please can you tell me what this item is?

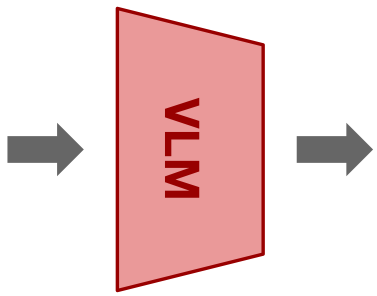


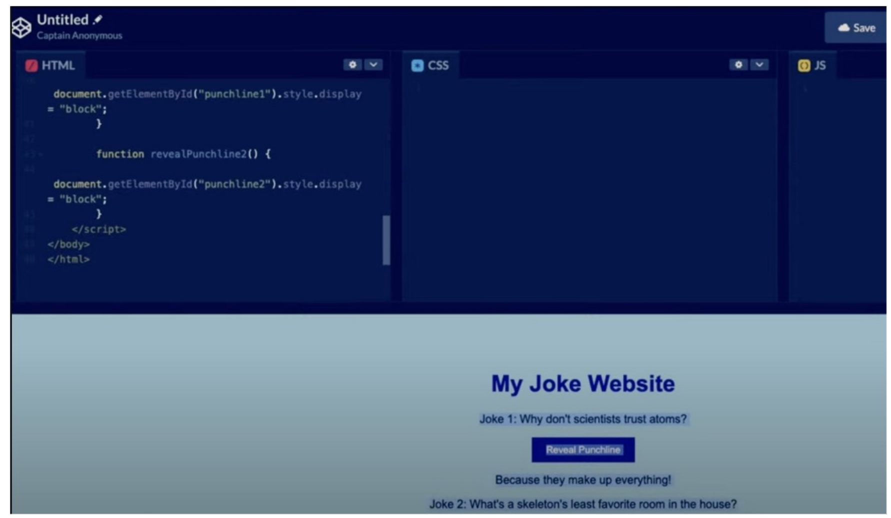
A: butternut squash red pepper soup

VizWiz (VizWiz VQA)

Image-to-Text Applications: Sketch-to-Website







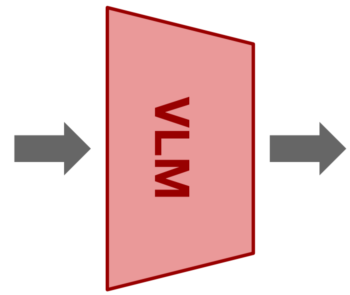
GPT-4 Demo (twitter.com)

Image-to-Text Applications: Image to Latex

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}} = \frac{1}{1-2^{-s}} \cdot \frac{1}{1-3^{-s}} \cdot \frac{1}{1-5^{-s}} \cdot \frac{1}{1-7^{-s}} \cdots$$

\+

Can you write this in LaTeX?




```
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\cdots$
```
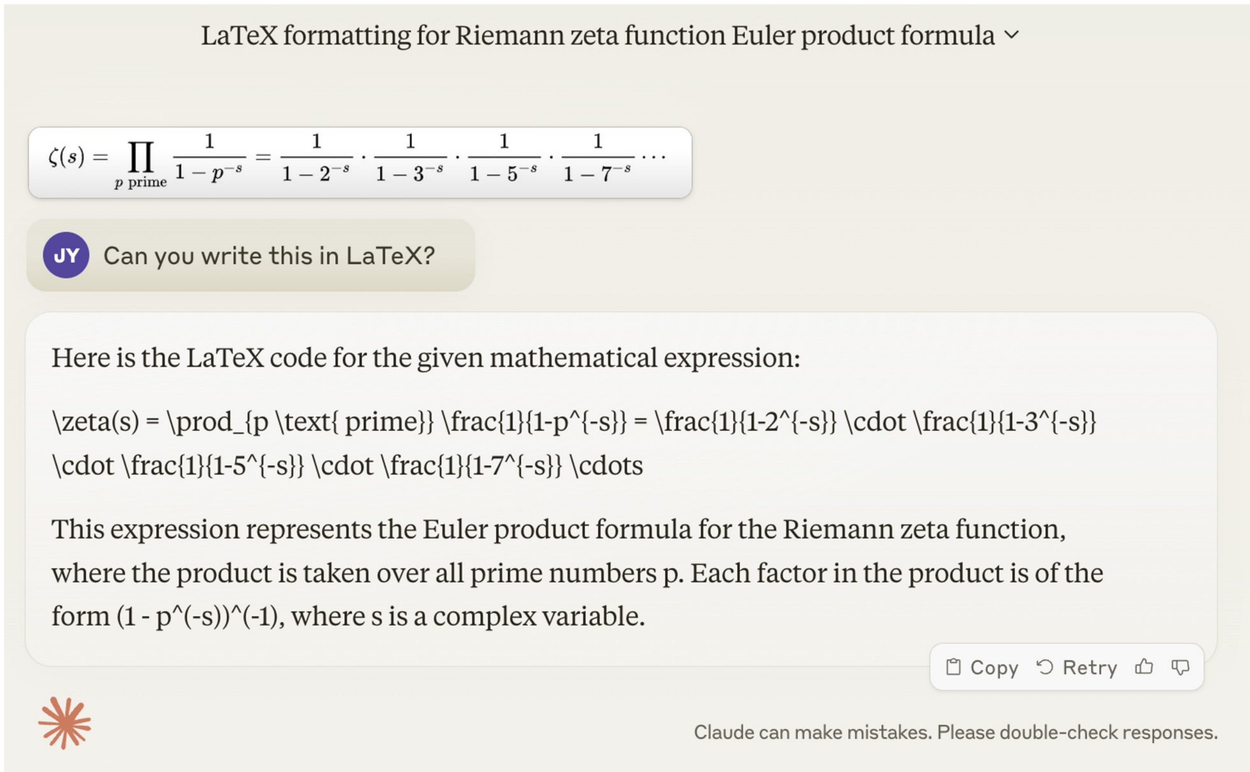

Image-to-Text Applications: Image to Latex

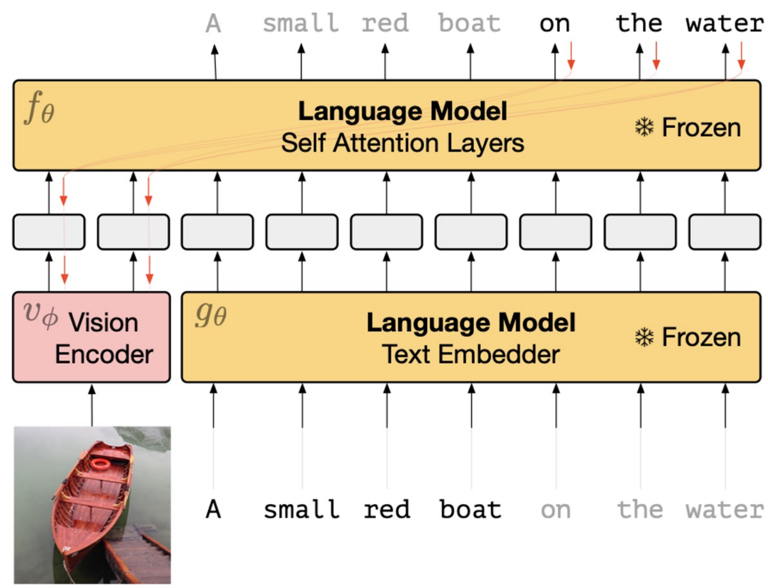


Claude (claude.ai)

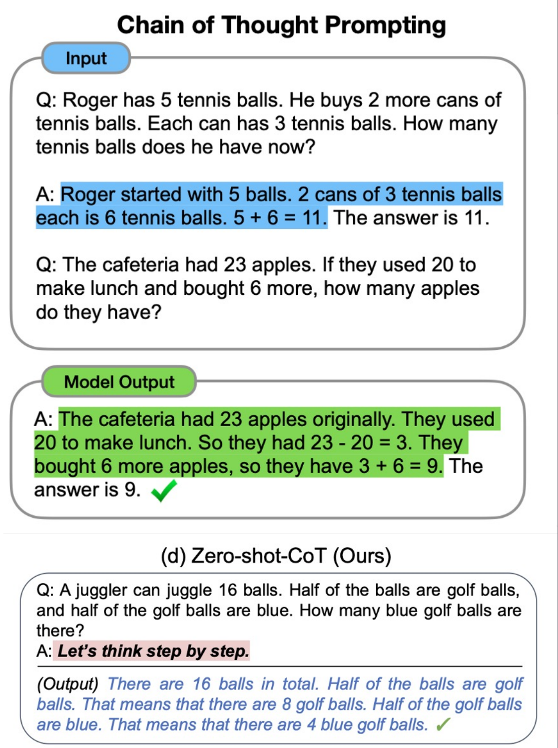
How do you train VLMs?

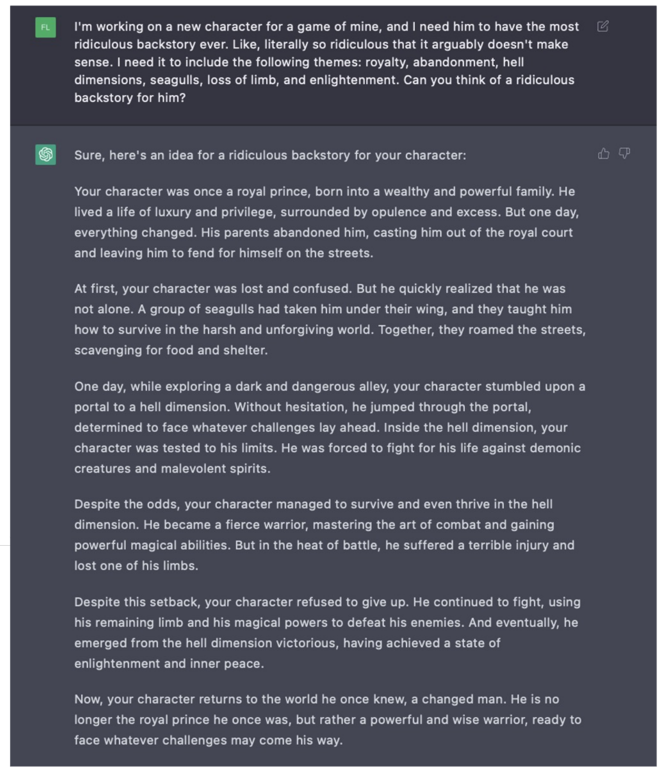
Training Multimodal LLMs

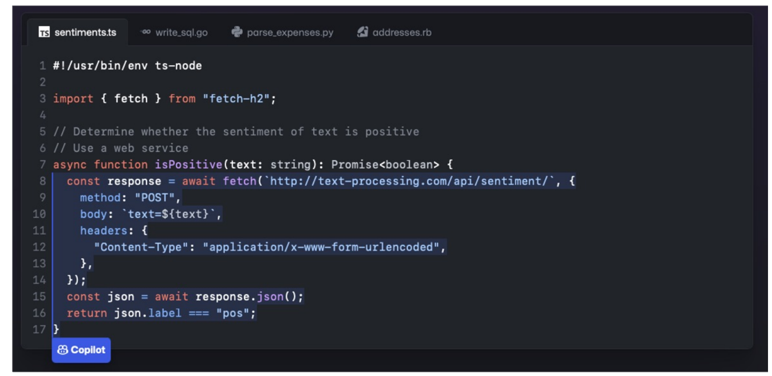
- Image + text inputs → text outputs
  - Very general!
- **Early on (2014 - 2020):** Finetune on paired image-text datasets for specific tasks
- **More recently:** Staple a pretrained vision model to a pretrained LLM

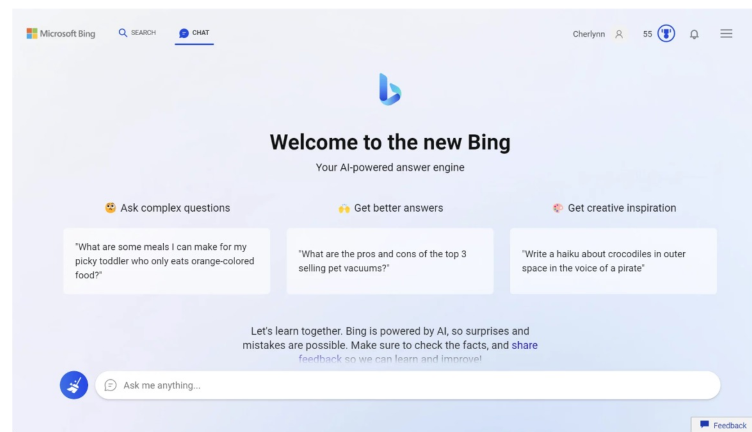


LLMs are impressive general models





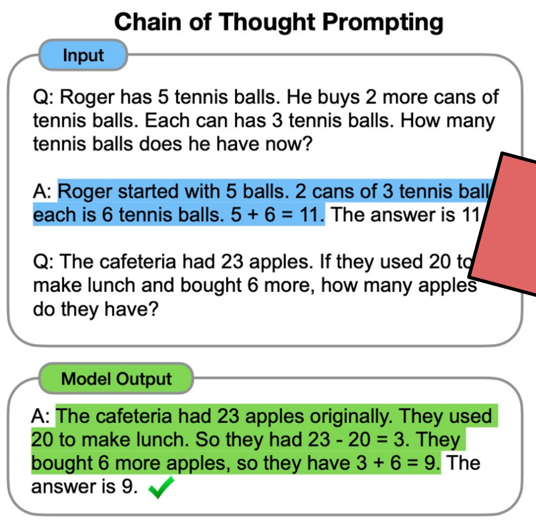


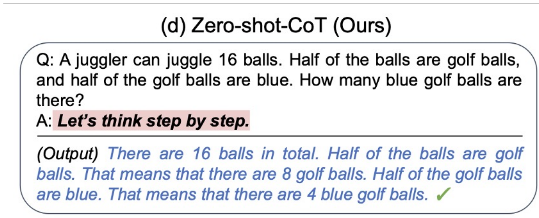


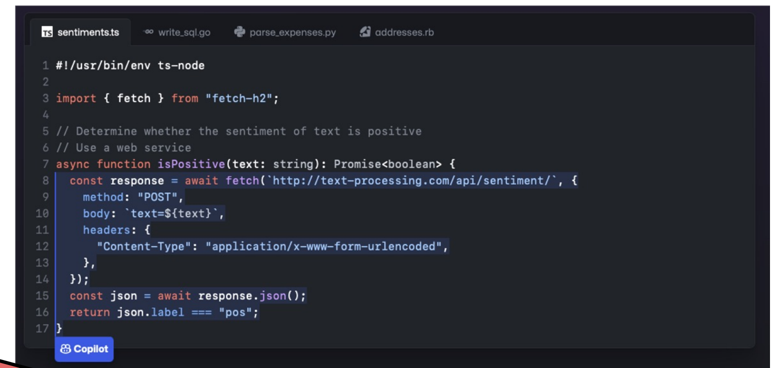
Chain of Thought Prompting Elicits Reasoning in Large Language Models (Wei et al., 2022)
Large Language Models are Zero-Shot Reasoners (Kojima et al., 2022)
ChatGPT (OpenAI, 2022)
Copilot (GitHub, 2021)
Bing (Microsoft, 2023)

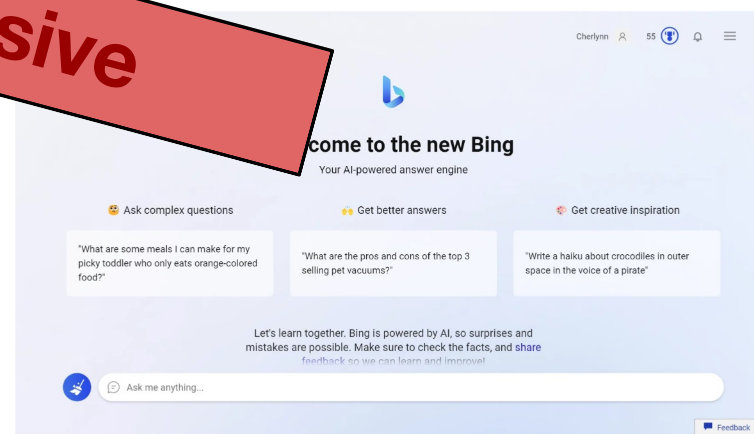
LLMs are impressive general models

Resource Intensive









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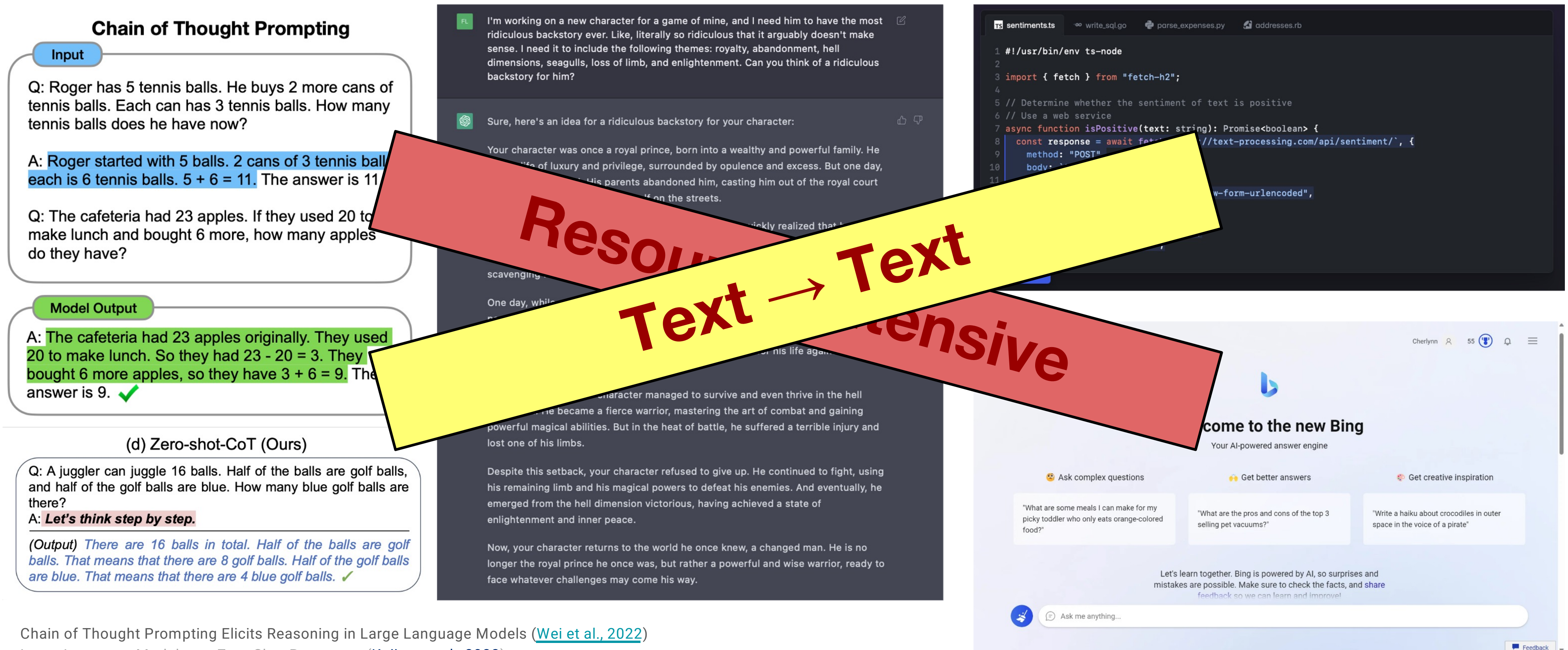

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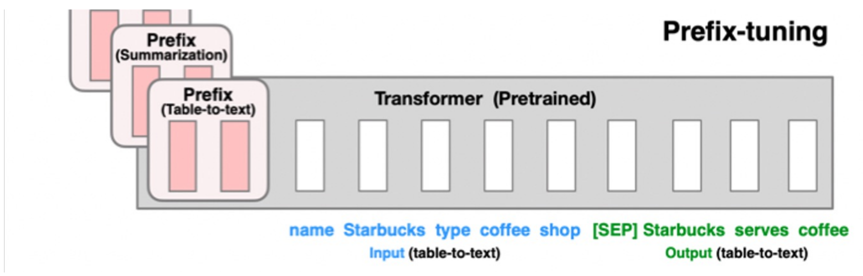
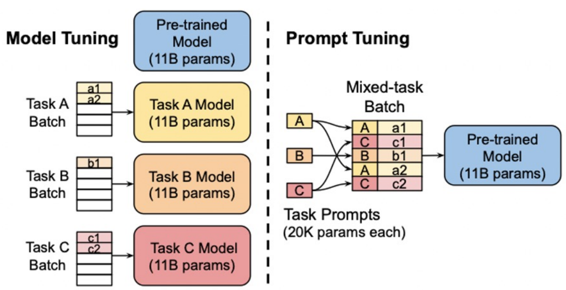
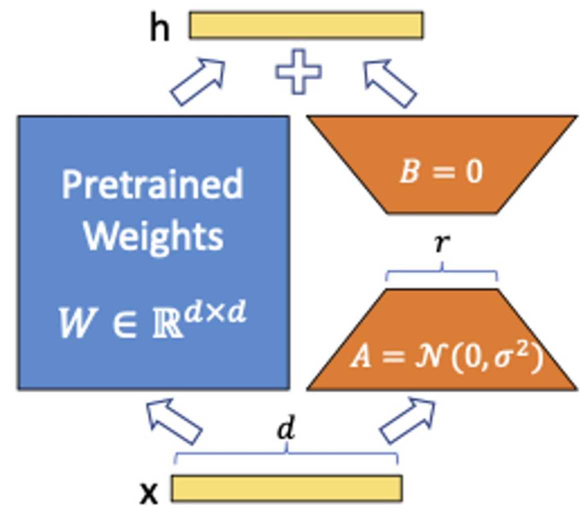
Parameter Efficient Adaptation



Prefix Tuning: Learns a prefix embedding (for each layer) to adapt to new tasks. ~99.9% of the model kept frozen.



Prompt Tuning: Similar idea to prefix-tuning, but learns just a single prefix for input embeddings.

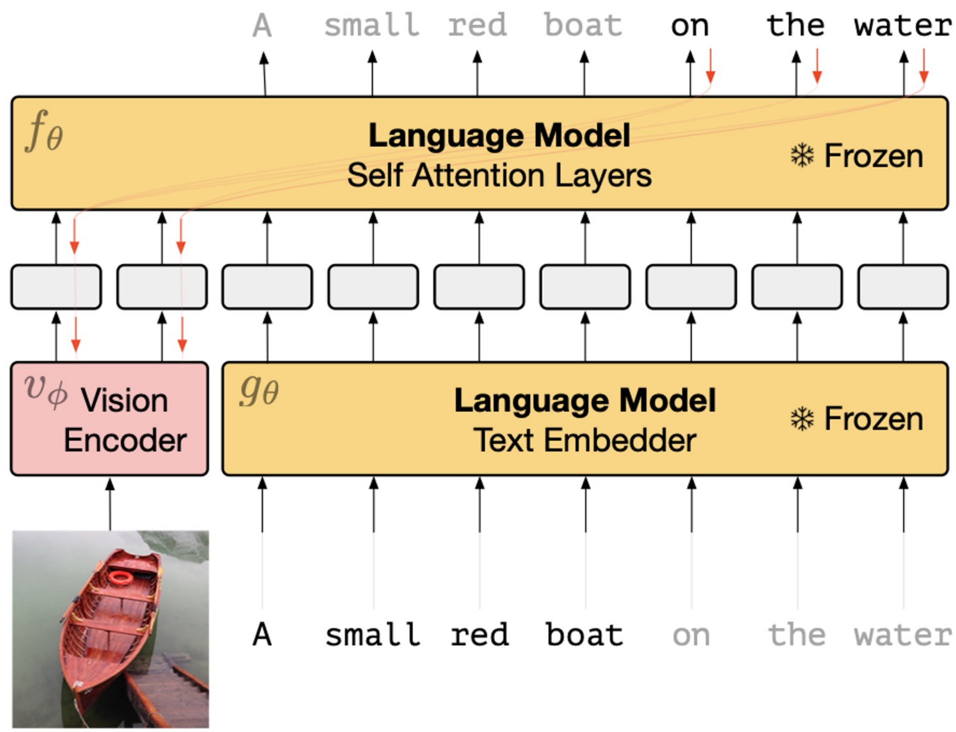


Low-Rank Adaptation (LoRA): Injects trainable rank decomposition matrices into each Transformer layer of a pretrained model.

Prefix-Tuning: Optimizing Continuous Prompts for Generation (Li and Liang, 2021)
The Power of Scale for Parameter-Efficient Prompt Tuning (Lester et al., 2021)
LoRA: Low-Rank Adaptation of Large Language Models (Hu et al., 2021)

Multimodal LMs: Frozen (2021)

- **Frozen**
  - Prefix tuning for adapting LLMs to image captioning
  - ~95% of the model kept frozen.
  - Capable of compelling few-shot multi-modal reasoning



Multimodal Few-Shot Learning with Frozen Language Models (Tsimpoukelli et al., 2021)

Multimodal LMs: Frozen (2021)

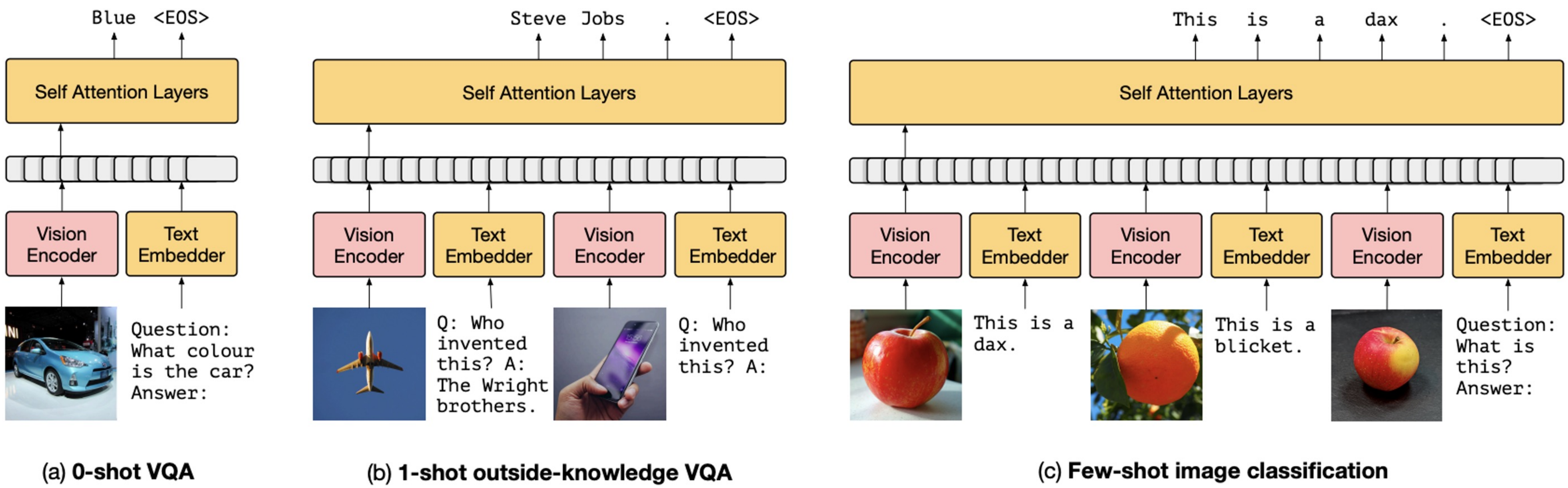


Figure 3: Inference-Time interface for *Frozen*. The figure demonstrates how we can support (a) visual question answering, (b) outside-knowledge question answering and (c) few-shot image classification via in-context learning.

Multimodal Few-Shot Learning with Frozen Language Models (Tsimpoukelli et al., 2021)

Frozen model Mapping Loss

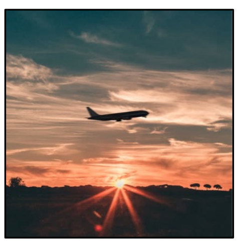


Image #1

silhouette of a plane against the sunset

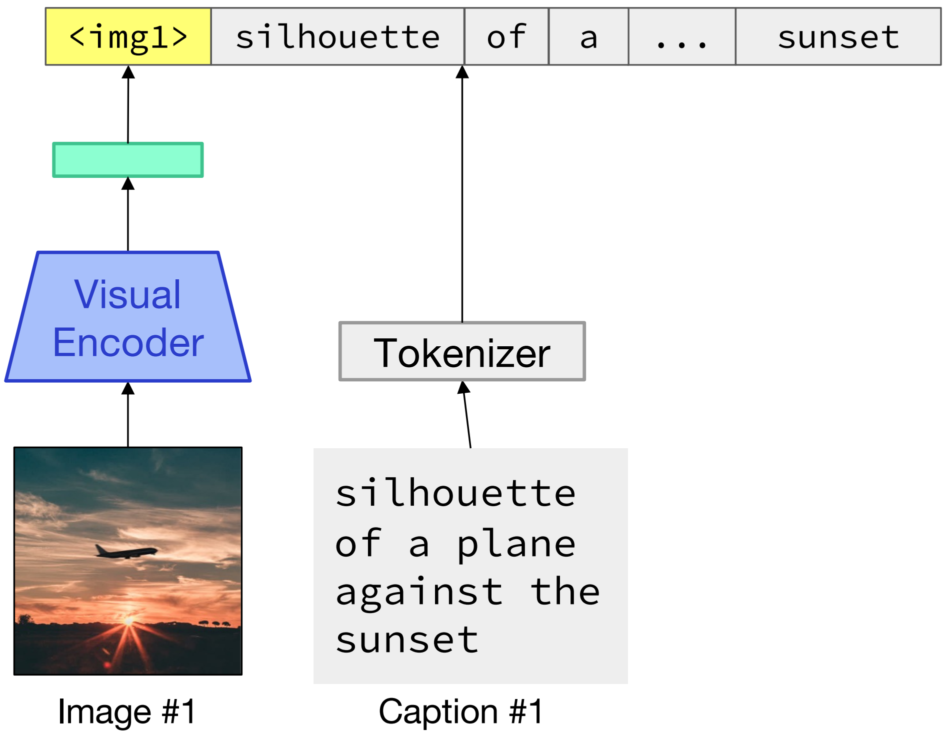
Caption #1

Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)

Frozen model Mapping Loss

Input Embeddings

(seq_len, 4096)



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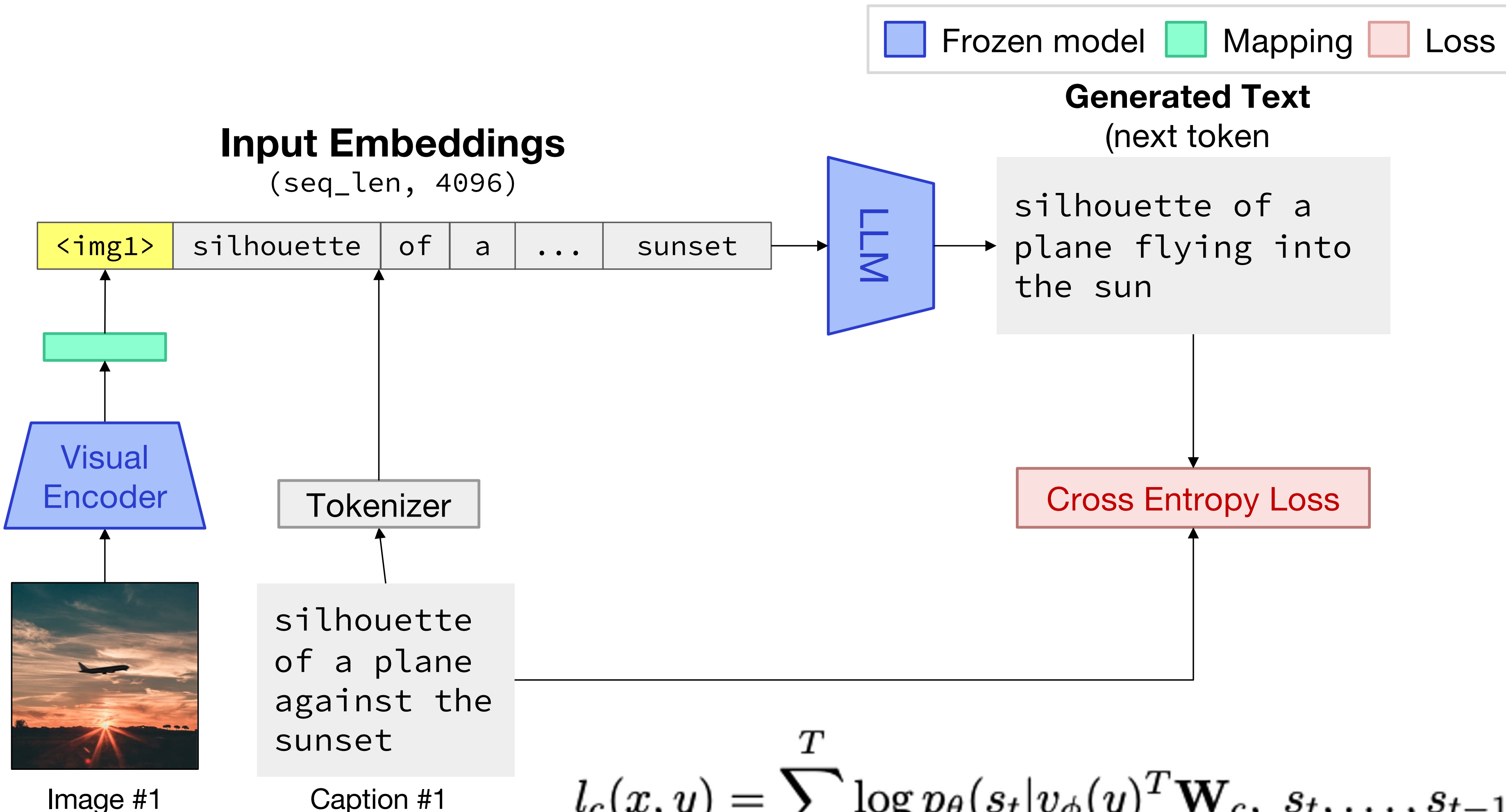
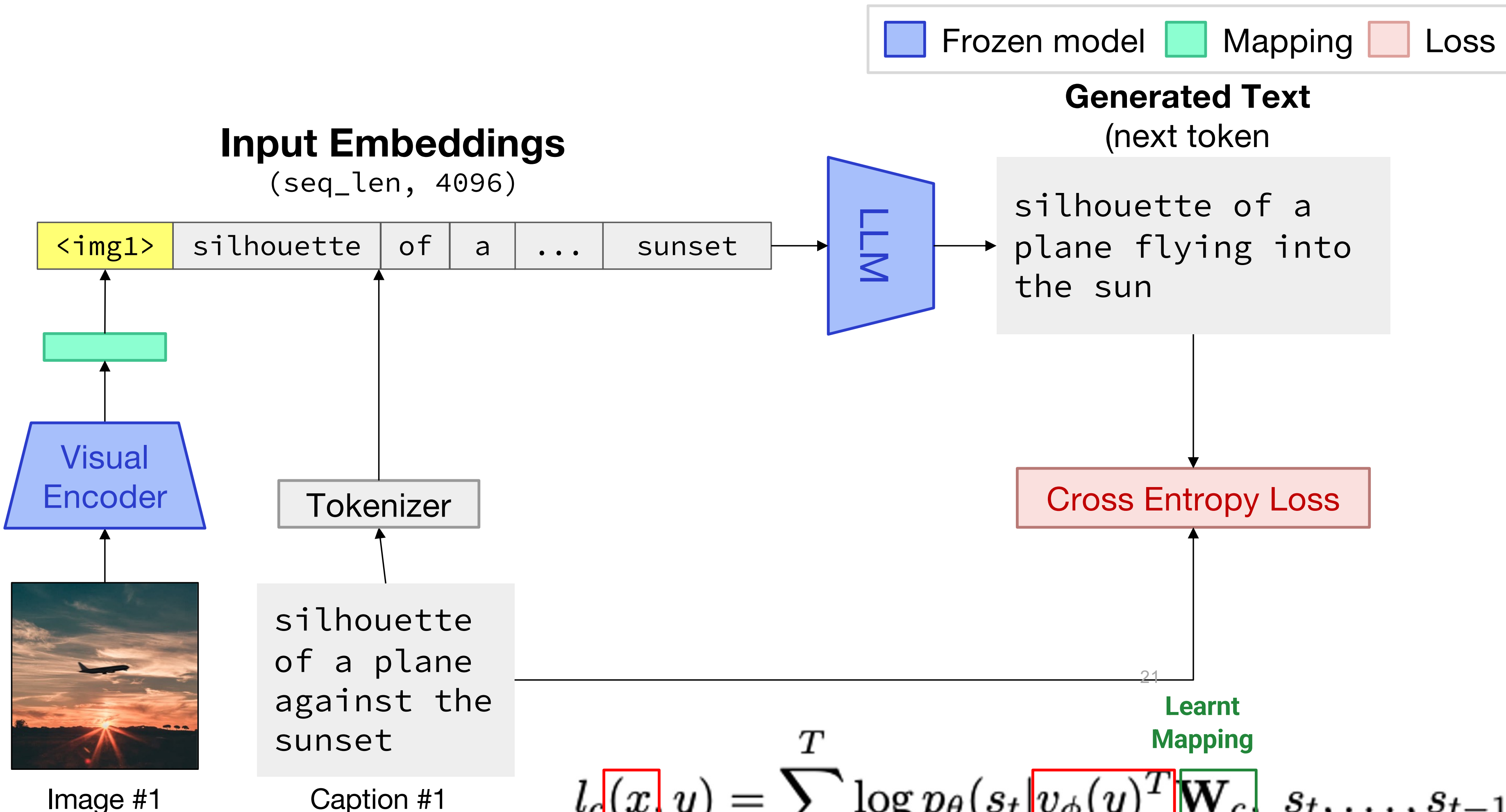


$$l_c(x, y) = \sum_{t=1}^{T} \log p_\theta(s_t | v_\phi(y)^T \mathbf{W}_c,\ s_t, \ldots, s_{t-1})$$

Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)



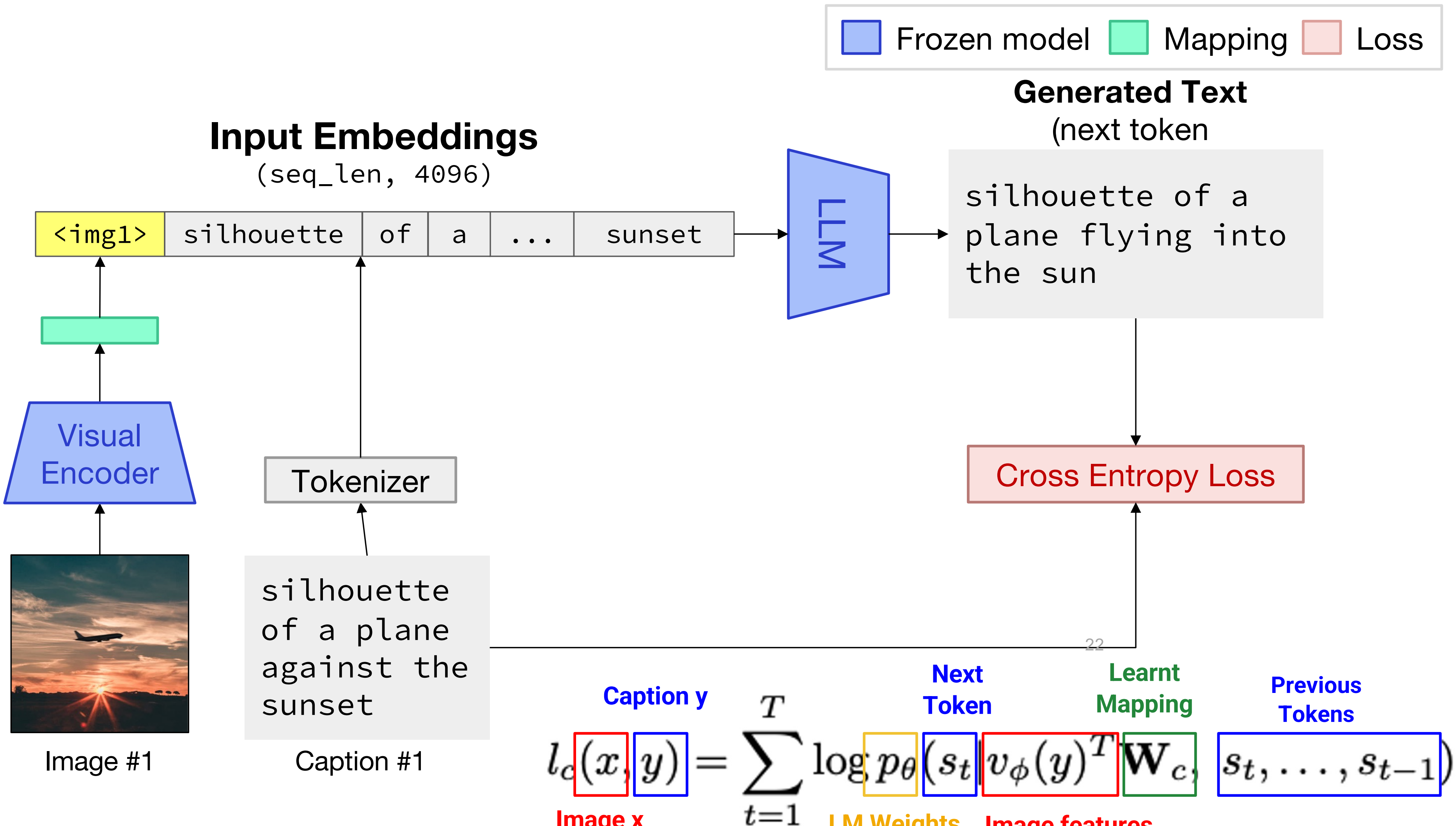
Learnt Mapping

$$l_c(\boxed{x}, y) = \sum_{t=1}^{T} \log p_\theta(s_t | \boxed{v_\phi(y)^T} \boxed{\mathbf{W}_c}, s_t, \ldots, s_{t-1})$$

Image x

Image features

Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)



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Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)

How visually grounded are text-only LLMs?

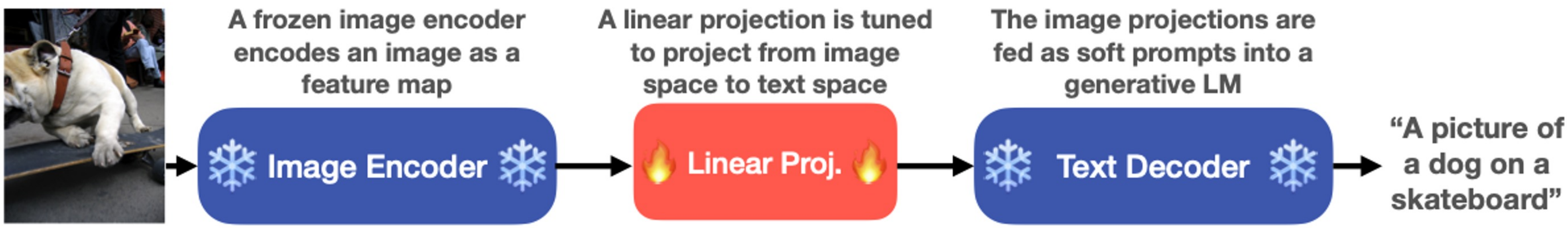
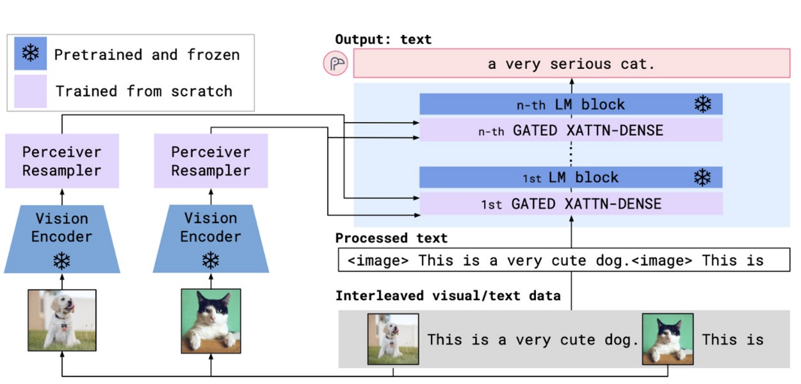


Figure 1: We train linear projections from image representations into the input space of a language model to produce captions describing images. We find that LMs can describe the contents of most image representations, but performance varies based on the type of image encoder used.

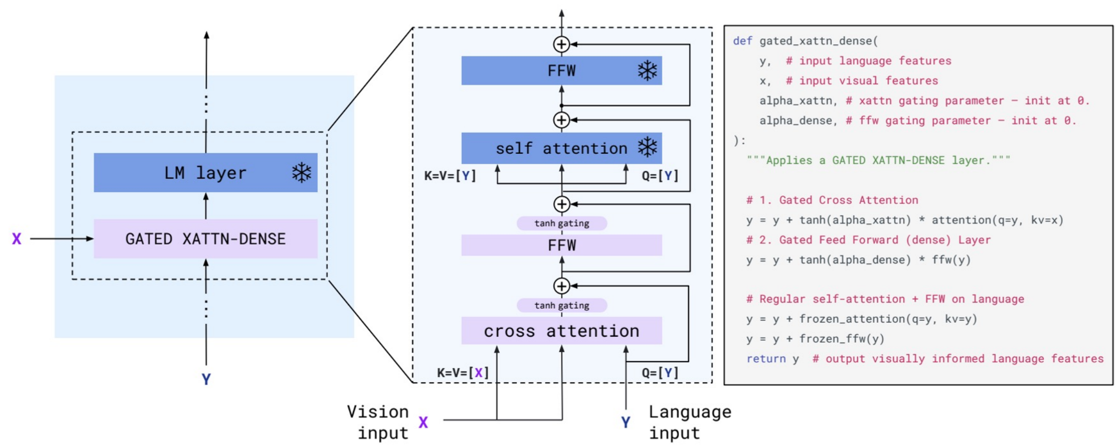
Merullo et al. showed that pretrained text-only LMs and pretrained visual encoders produce <u>functionally equivalent representations up to a linear mapping</u>.

Linearly Mapping from Image to Text Space ([Merullo et al., 2022](#))

Multimodal LMs: Flamingo

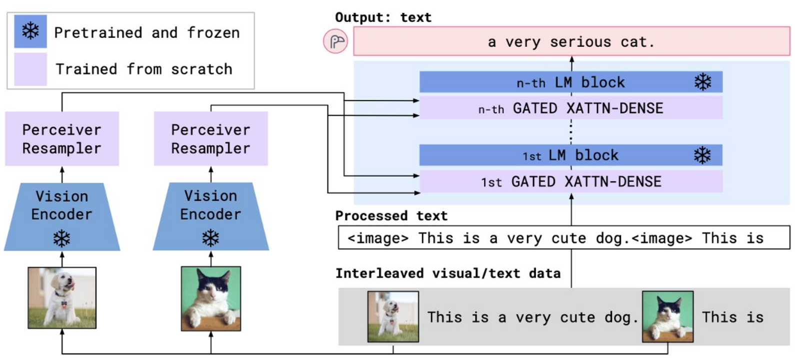


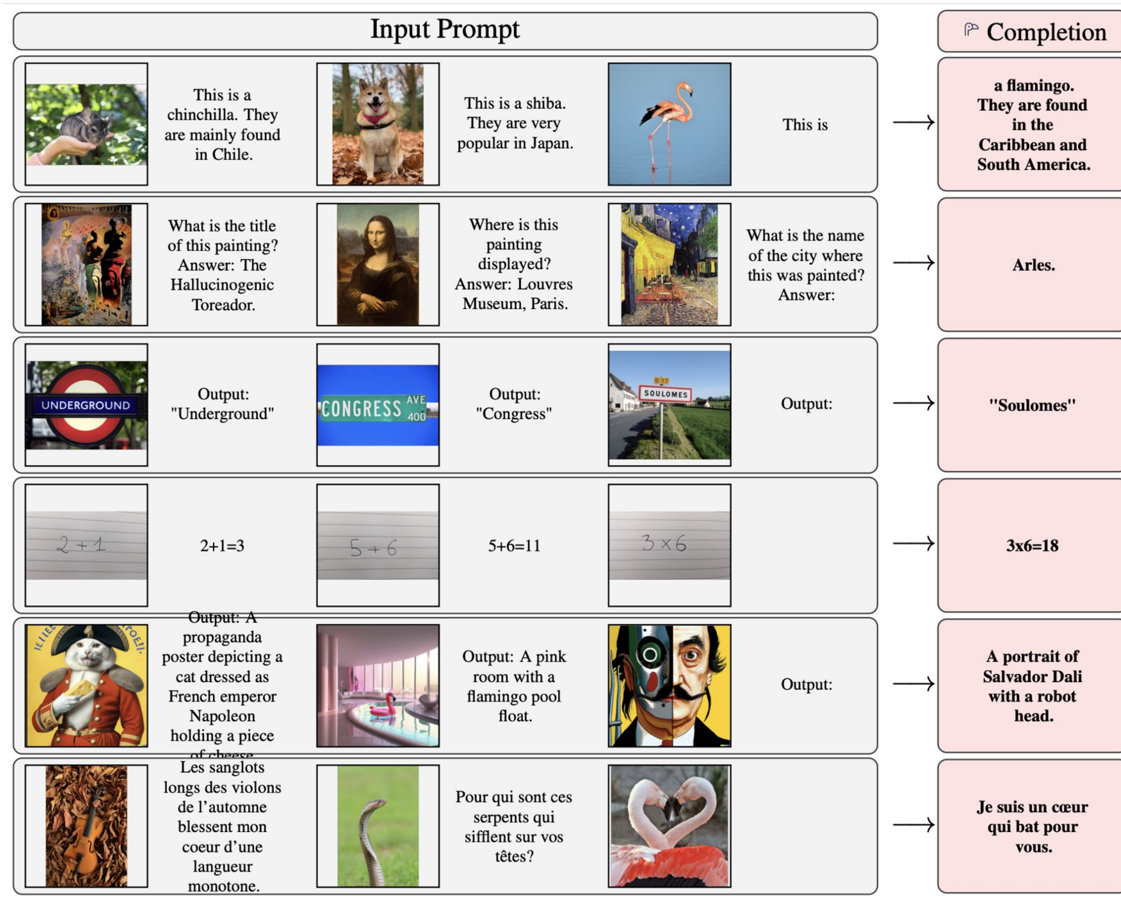
Flamingo: Finetunes new cross-attention layers on top of a 70B LLM. Achieves SOTA on many multi-modal tasks.



Introduced cross-attention layers between existing frozen LLM layers. Purple blocks are finetuned, blue blocks are kept frozen.

Multimodal LMs: Flamingo





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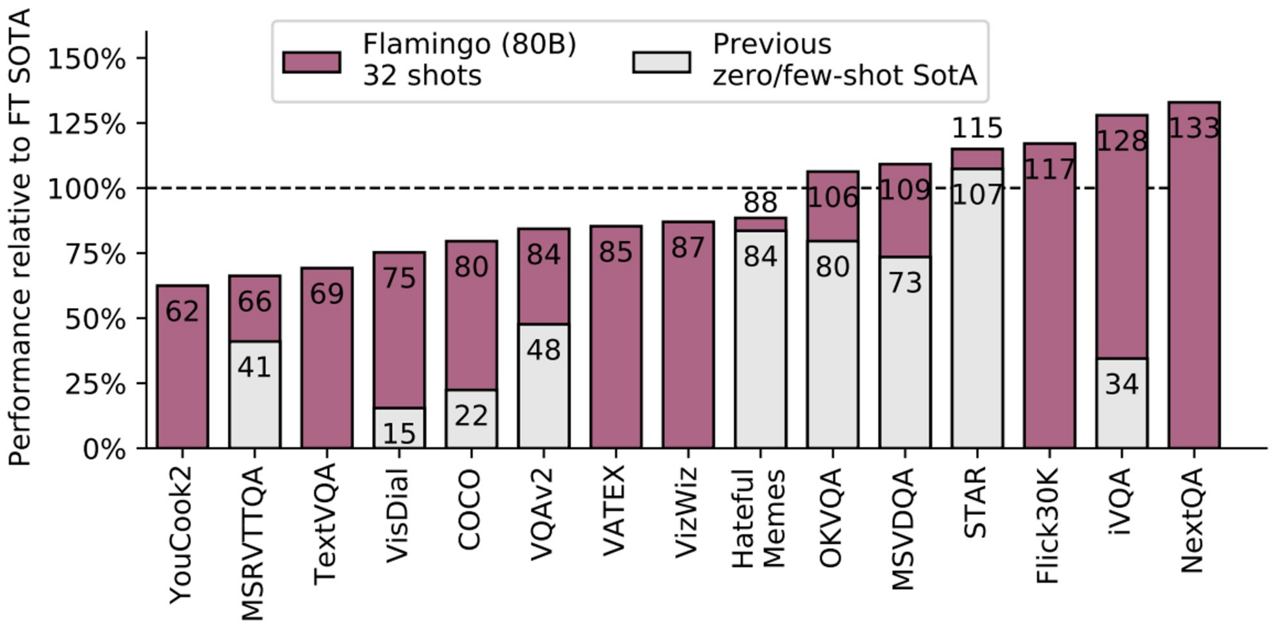
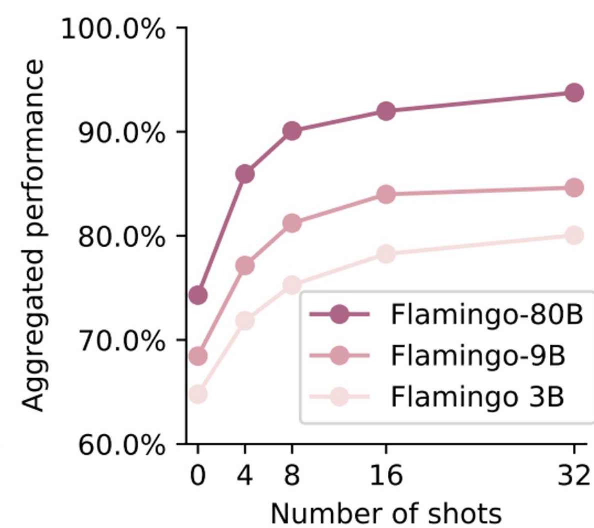
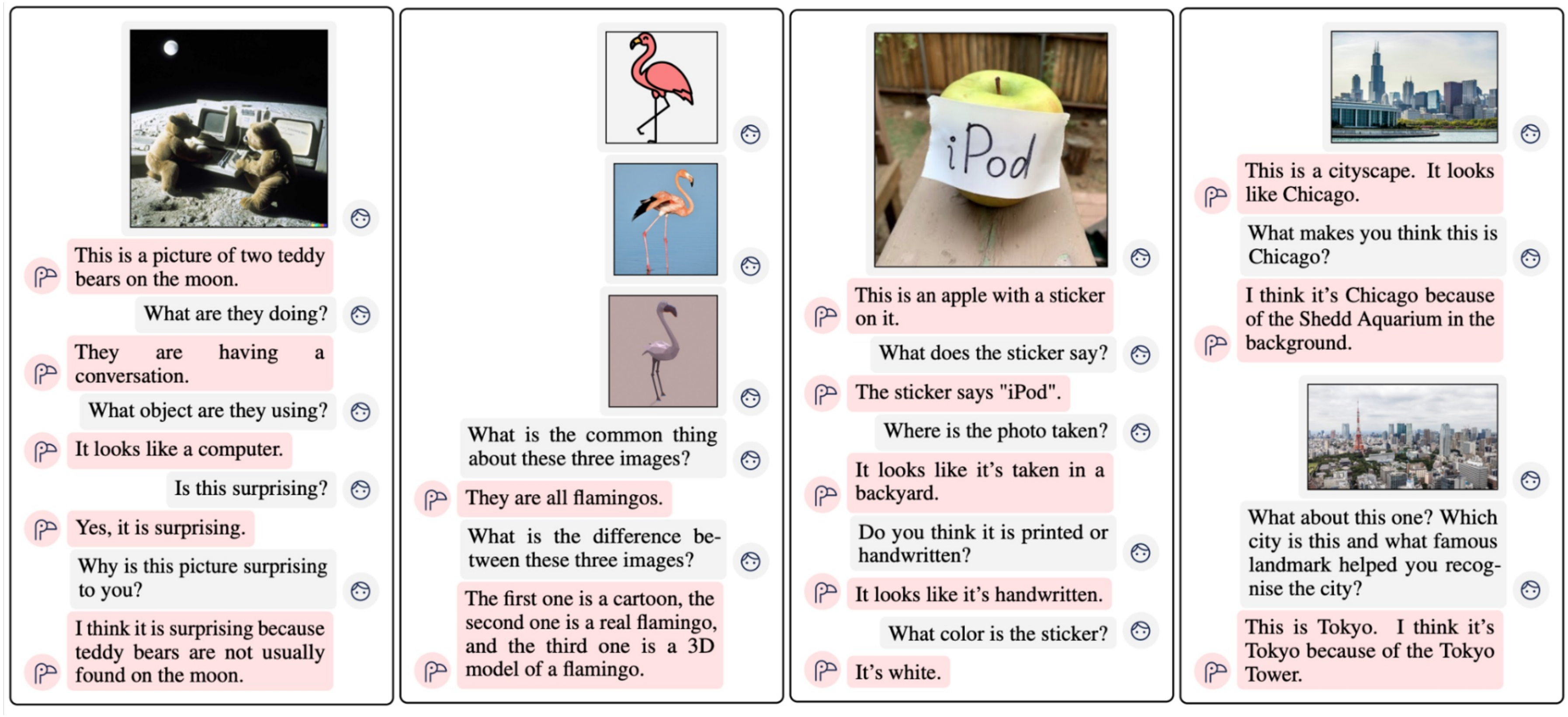
Multimodal LMs: Flamingo





Flamingo: a Visual Language Model for Few-Shot Learning (Alayrac et al., 2022)

Multimodal LMs: Flamingo



Flamingo: a Visual Language Model for Few-Shot Learning ([Alayrac et al., 2022](#))

Data: Flamingo

Image-Text Pairs dataset   Video-Text Pairs dataset   Multi-Modal Massive Web (M3W) dataset

Image-Text Pairs:
1.8B pairs with short captions (~12 tokens each)
312M pairs with long descriptions (~20.5 tokens each)

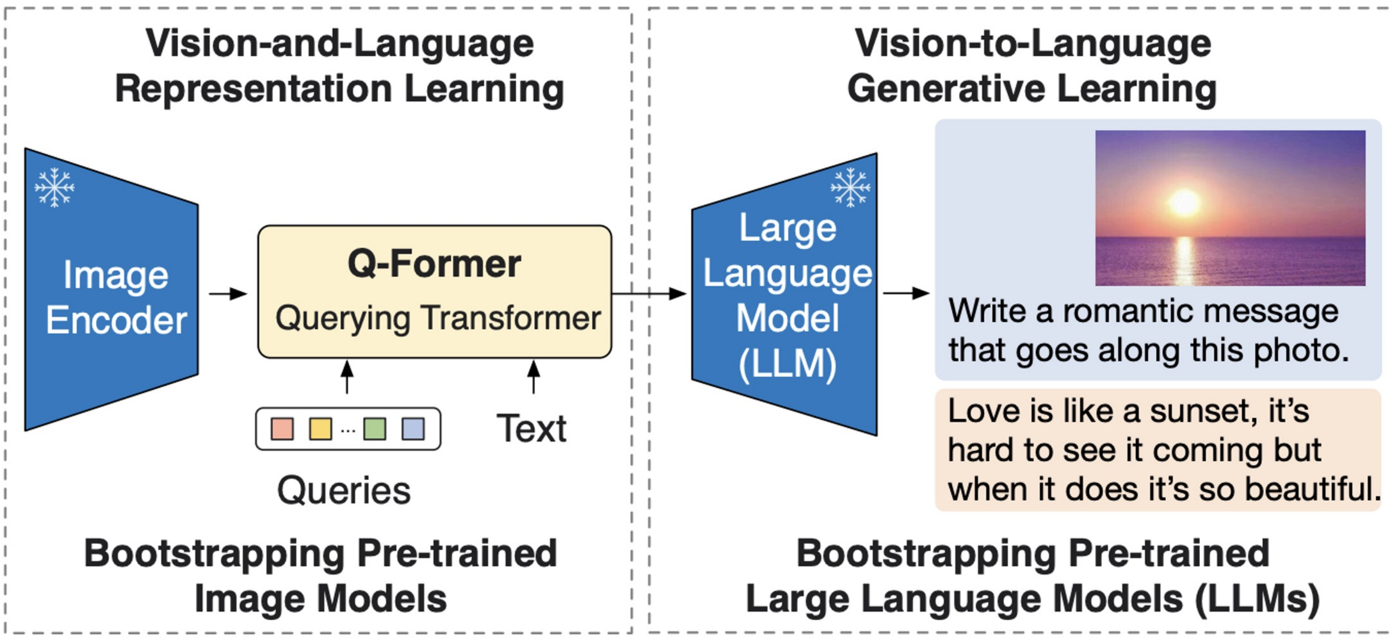
Video-Text Pairs: 27M short videos

28

M3W: 43M webpages (for each page, sample 256 tokens, take first 5 images)

Flamingo: a Visual Language Model for Few-Shot Learning (Alayrac et al., 2022)

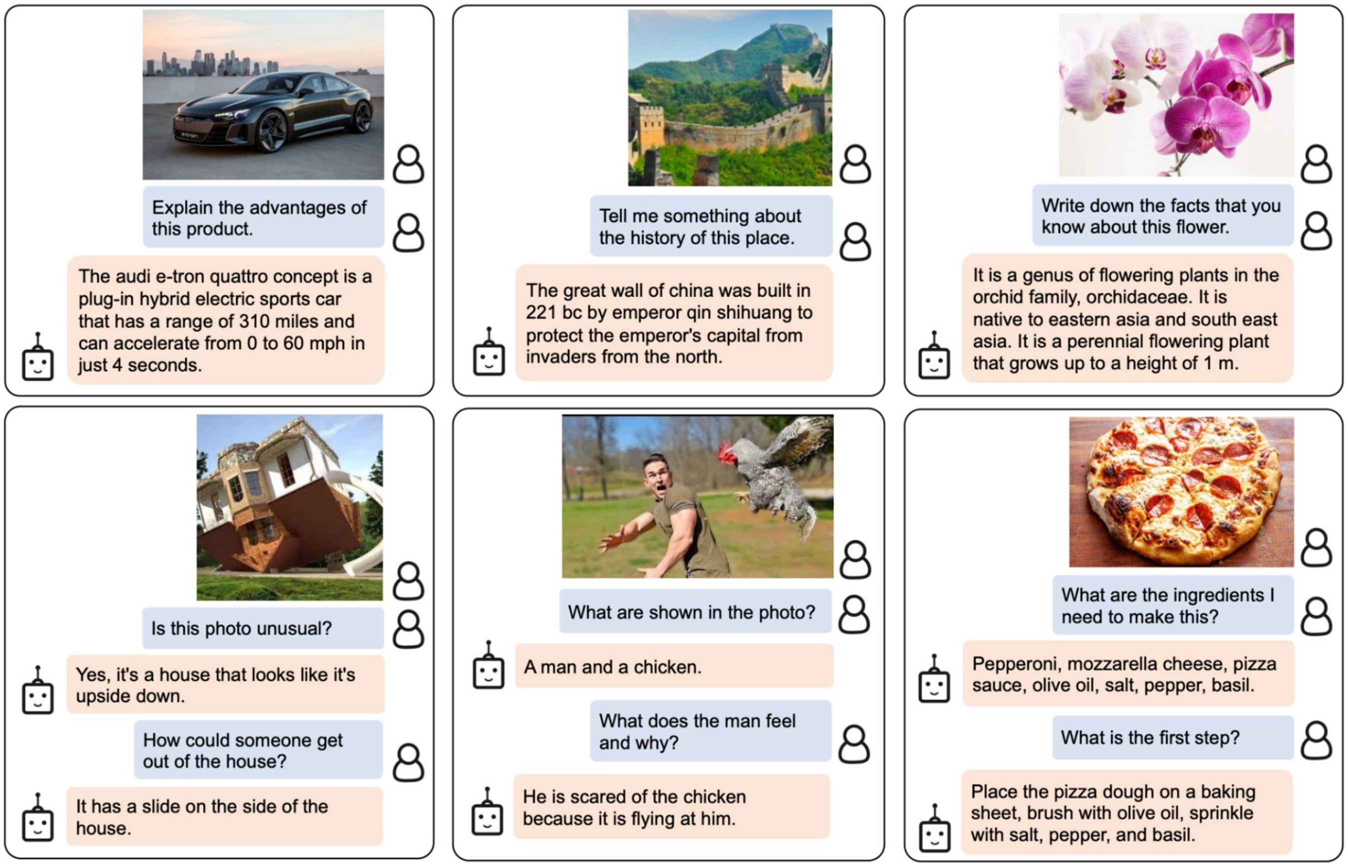
Multimodal LMs: BLIP-2



BLIP-2: Trains a Q-Former (essentially a transformer with learnt queries) to map between a frozen image encoder and a frozen LLM. Achieves strong performance on image captioning, VQA, and other image-to-text tasks.

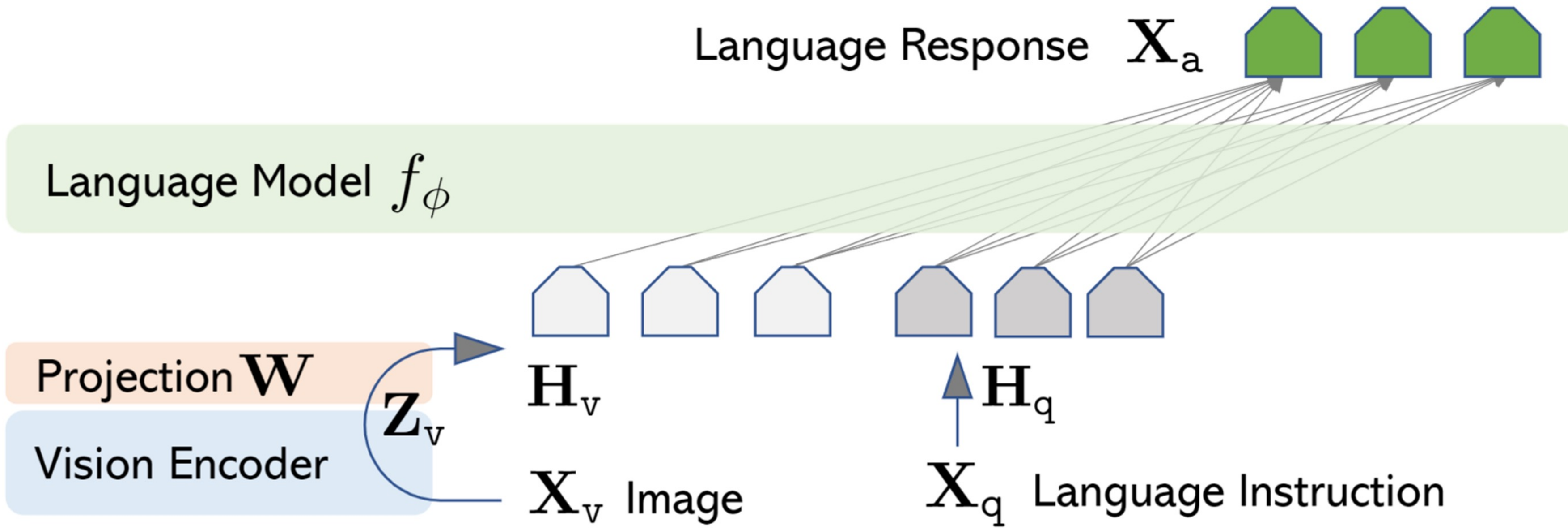
BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models ([Li et al., 2023](#))

Multimodal LMs: BLIP-2



BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models (Li et al., 2023)

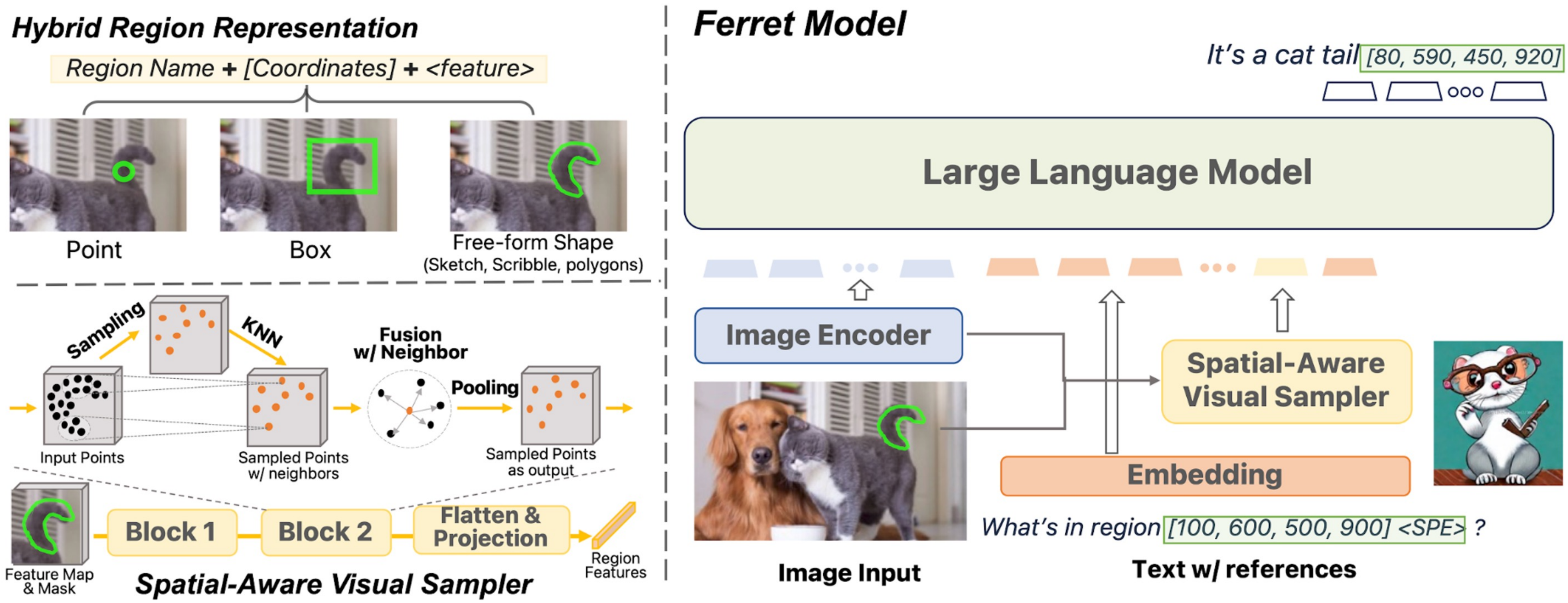
Multimodal LMs: LLaVA



LLaVA: Finetunes a linear layer (W) over a frozen vision encoder and a frozen LLM. Showcases strong performance by finetuning on paired data of images and text instructions (some GPT-4 generated).

Visual Instruction Tuning (Liu et al., 2023)

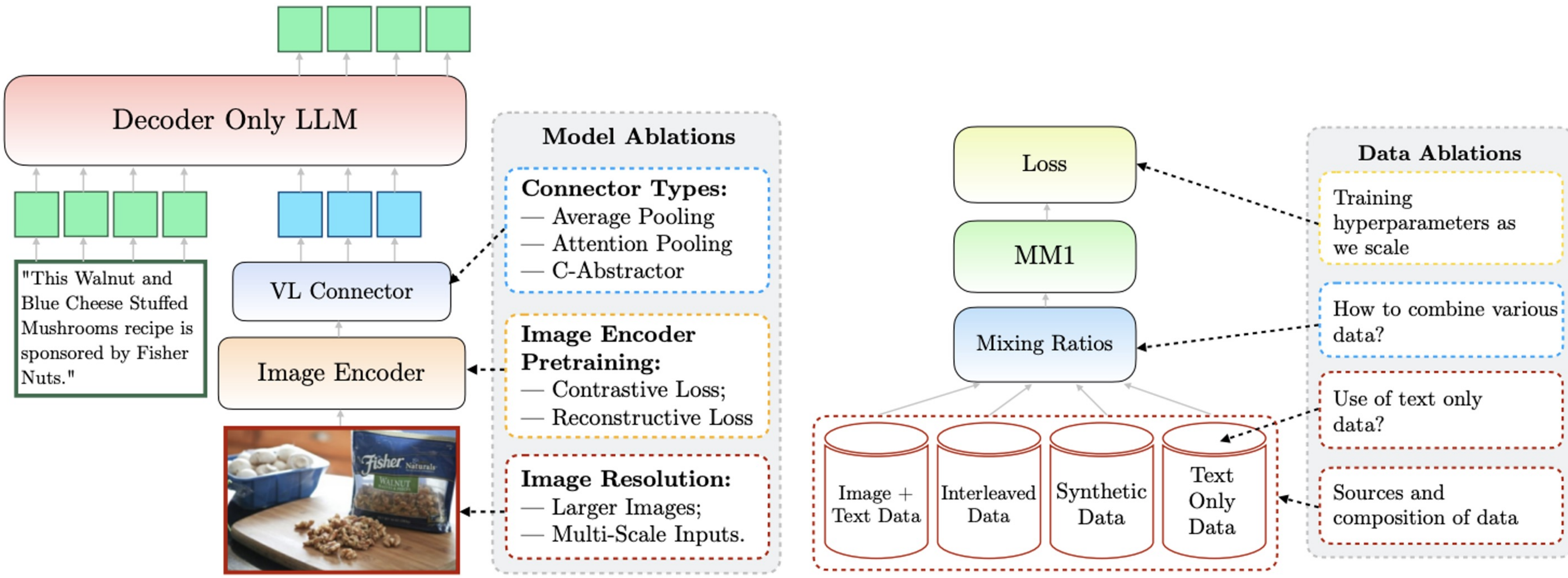
Multimodal LMs: Ferret



Ferret: Hybrid input representations, enabling fine-grained grounding and referencing.

An End-to-End MLLM that Accept Any-Form Referring and Ground Anything in Response (You et al., 2023)

Multimodal LMs: MM1



MM1: ViT-L image encoder + C-Abstractor (CNN) mapper + 1.2B decoder-only LM

MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training (McKinzie et al., 2024)

Multimodal LMs: MM1 Analysis

- **Biggest impact on results**:
  - Input image resolution has the highest impact (224px → 336px leads to +3% boost)
  - Model size (+1%) and training data composition (+1%) are also impactful

- **Vision-Language Connector**:
  - Number of visual tokens and image resolution matters most
  - The particular architecture has little effect

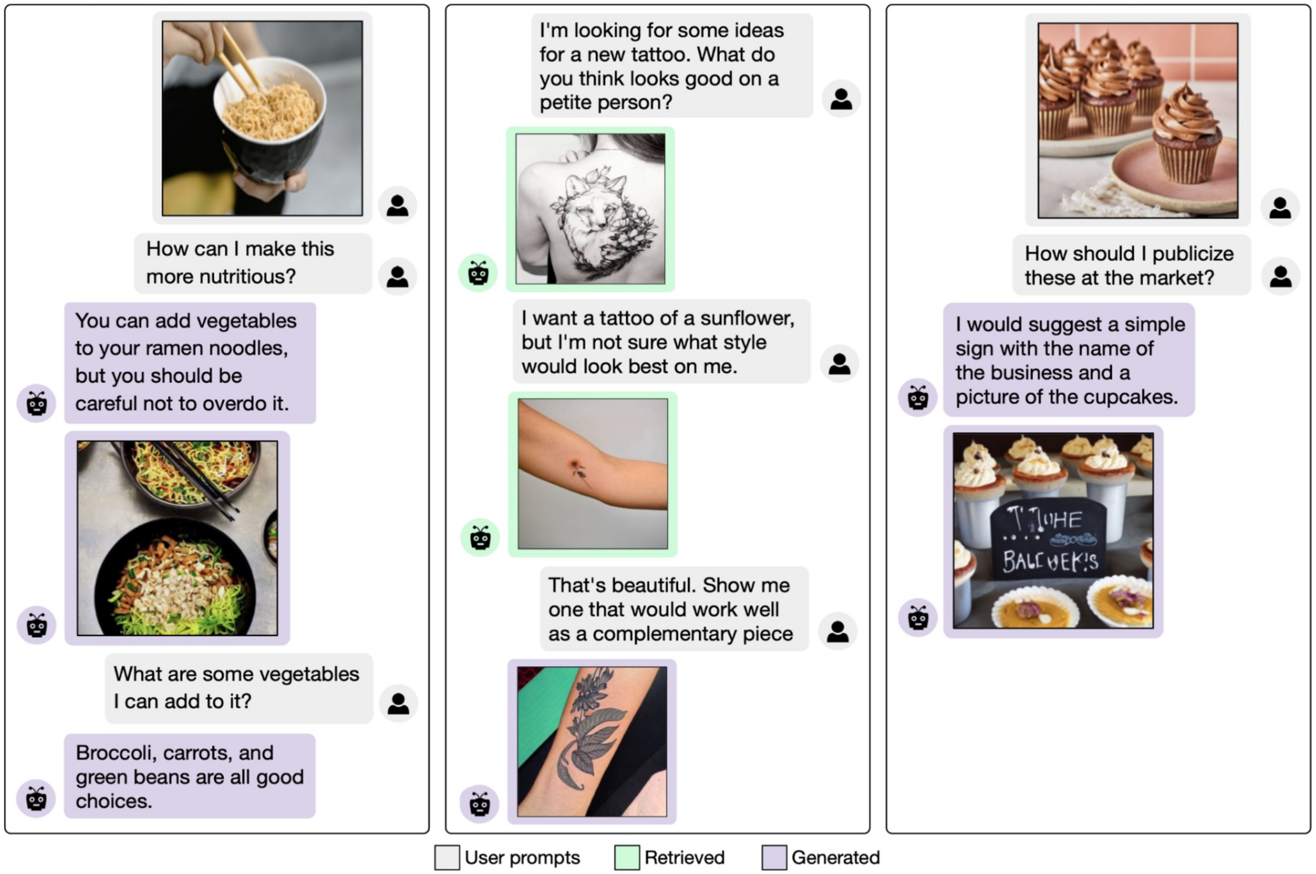
- **Pretraining Data**:
  - Interleaved data is instrumental for few-shot and text-only performance
  - Caption data lifts zero-shot performance
  - Text-only data helps with few-shot and text-only performance
  - Synthetic captions help with few-shot learning

Multimodal LMs: Others

- **Closed models:** GPT-5, Gemini, Claude 4
- **Open sourced models:**
  - Llama Family
  - Gemma
  - OpenFlamingo
  - Qwen-VL
  - CogVLM
  - and more each day…

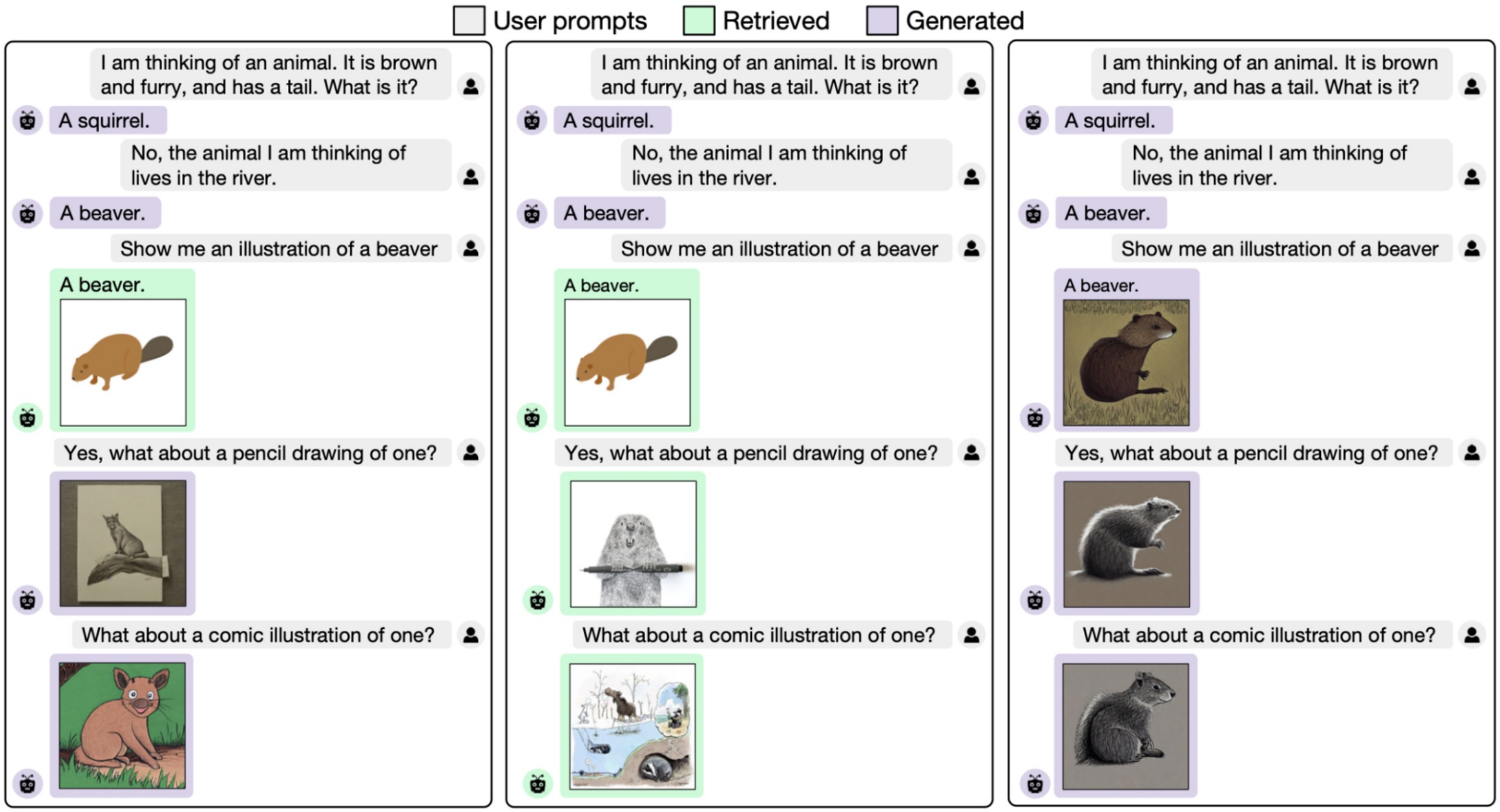
Can we ground text-only LLMs to consume and produce visual data?

GILL: Generating Images with Large Language Models



GILL reference

GILL: A More General Multimodal LM



Ours
Retrieval and generation multimodal LM. Decides when to retrieve or generate.

FROMAGe
Retrieval only multimodal language model. Image outputs limited to the candidate retrieval set.

Stable Diffusion
Generation only text-to-image model. Less sensitive to longer text inputs (such as dialogue).

GILL: A More General Multimodal LM

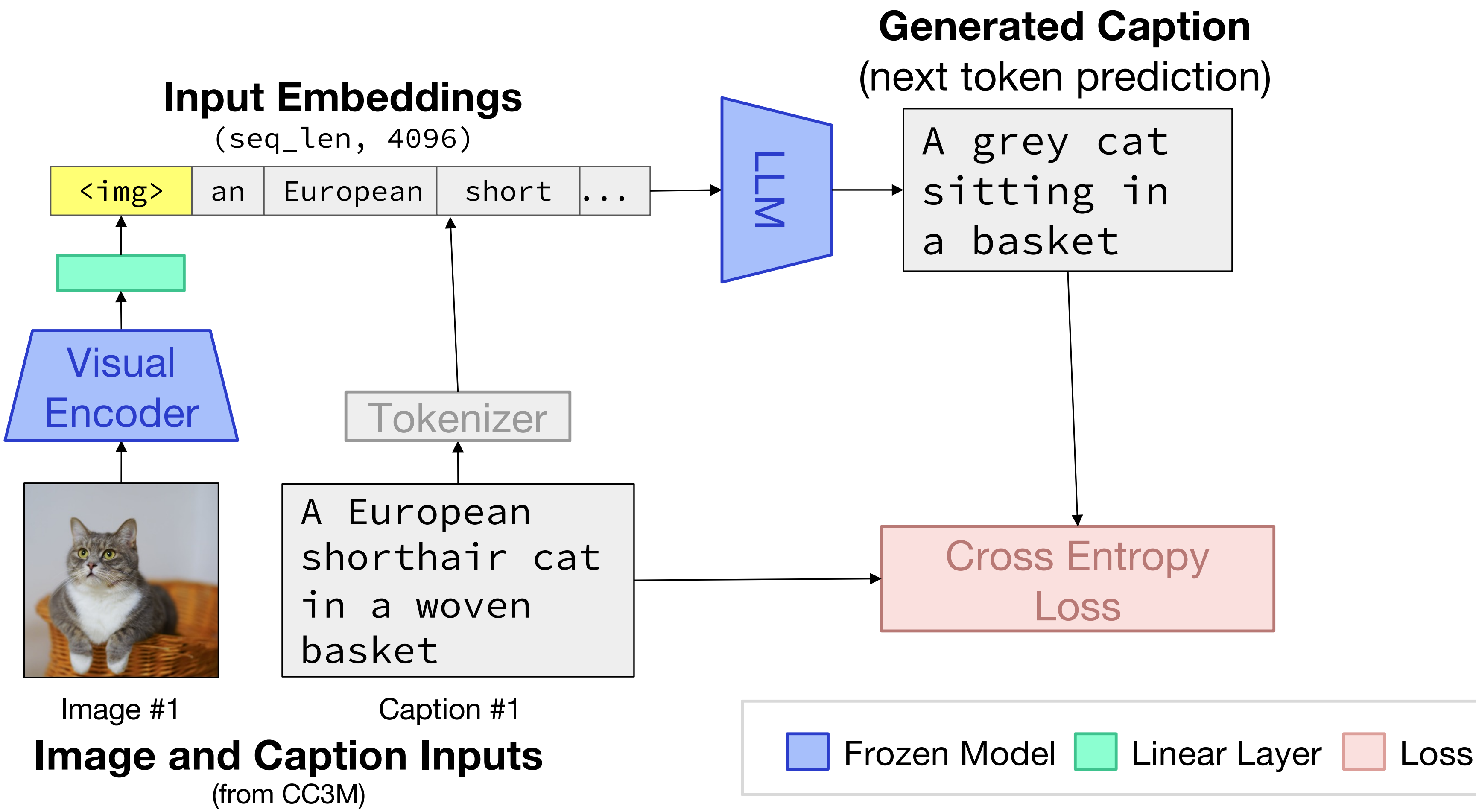
- **Frozen** (Tsimpoukelli et al., 2021)
  Flamingo (Alayrac et al., 2022)
  BLIP-2 (Li et al., 2023)
  - Process image + text, generate text only
- **FROMAGe** (Koh et al., 2023)
  - Process image + text, generate text + retrieve images
- **GILL**
  - Process image + text, generate text + retrieve images + generate images
  - Decides whether to retrieve images or generate from scratch
  - Resource efficient: trained on 2 GPUs for 2 days

Multimodal Few-Shot Learning with Frozen Language Models (Tsimpoukelli et al., 2021)
Linearly Mapping from Image to Text Space (Merullo et al., 2023)
Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)

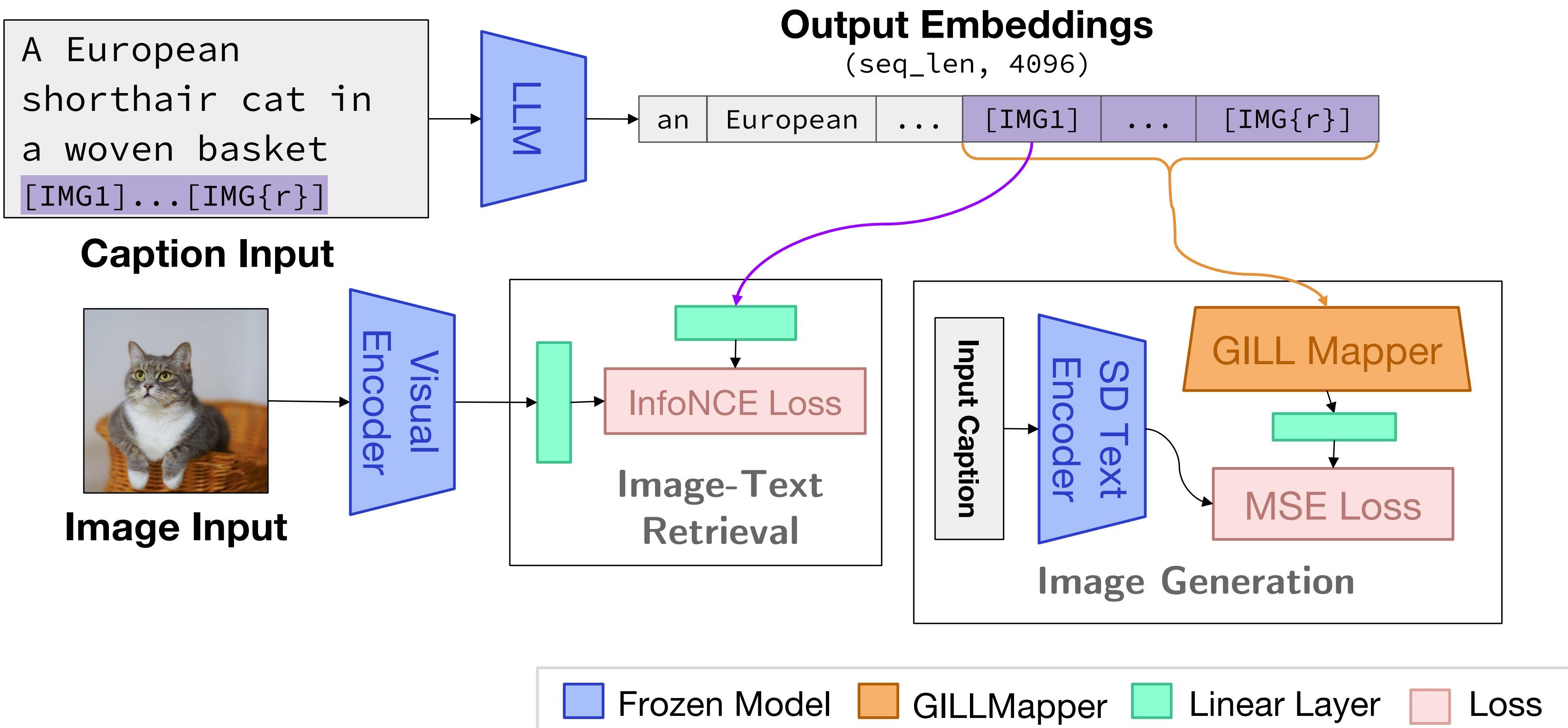
GILL: A More General Multimodal LM

- **Capable of retrieving images, generating images, and generating text**
  - Can condition on arbitrarily interleaved image + text inputs
  - Generate text, generate images, and retrieve images as part of the output
- **Leverage the learnt abilities of pre-trained text-only LLMs**
  - In-context learning
  - Generate long and coherent dialogue
- **Model agnostic**
  - Use a 7B LLM, the CLIP encoder, and the Stable Diffusion image generator
  - Likely benefits from using larger and stronger LLMs in the future

Learning to Process Images

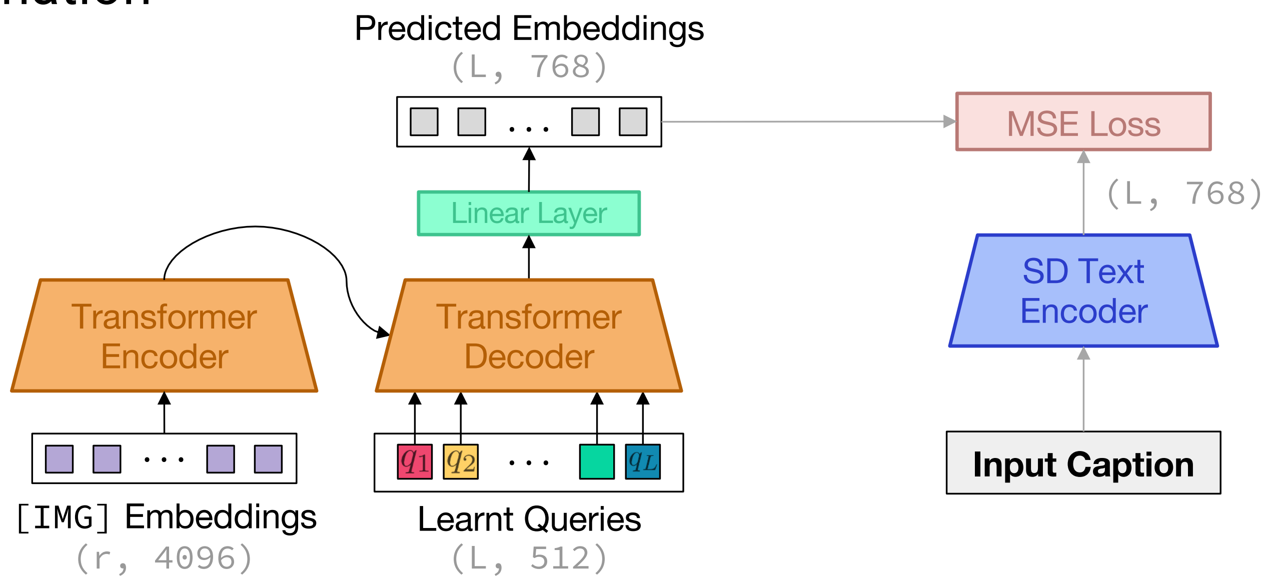


Learning to Produce Images



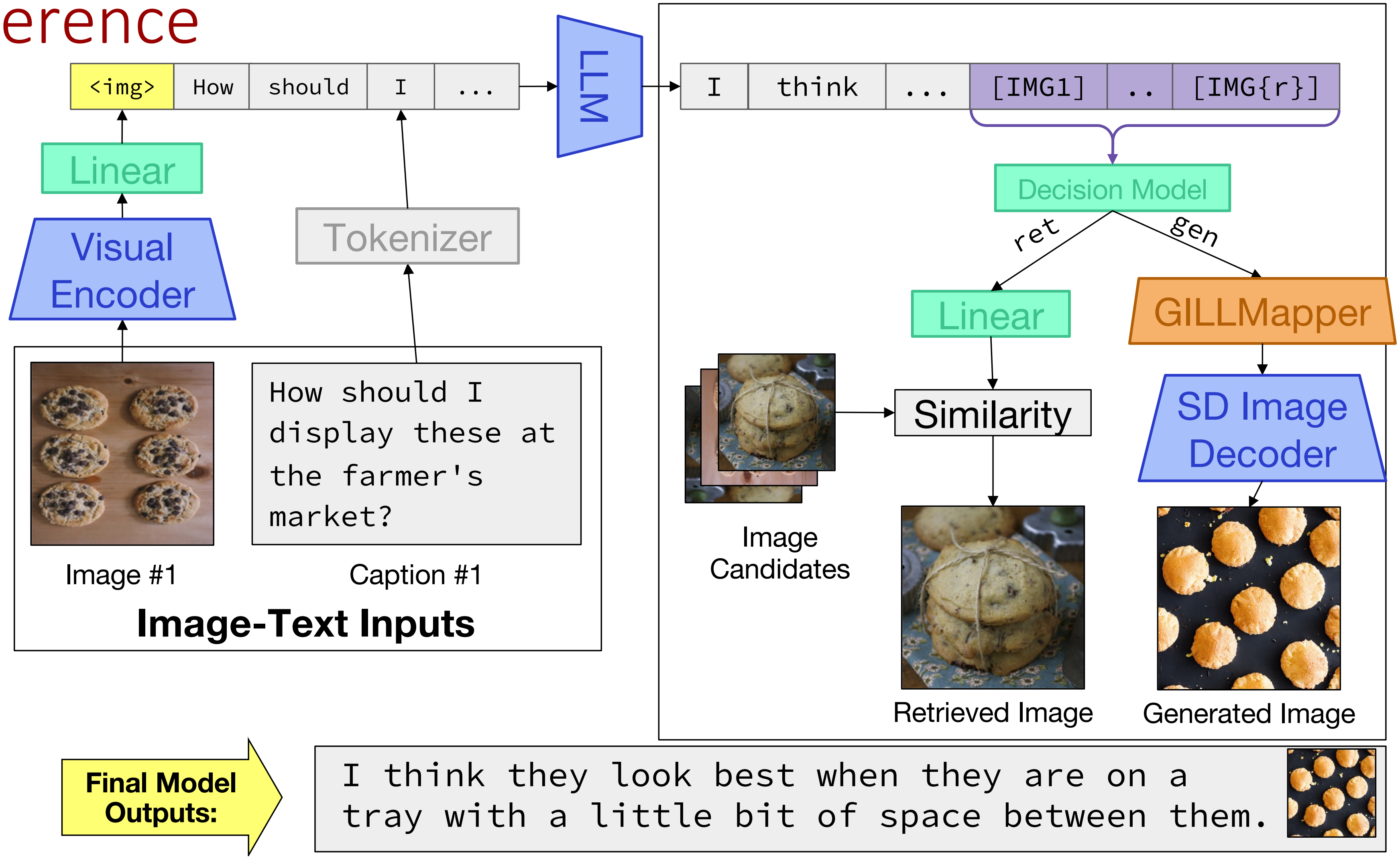
GILL Mapper: LLM-to-Generator Map

- Many approaches use linear mappings between LLMs and visual models
- This is insufficient for image generation: decoders require dense information



Multimodal Few-Shot Learning with Frozen Language Models (Tsimpoukelli et al., 2021)
Linearly Mapping from Image to Text Space (Merullo et al., 2023)
Grounding Language Models to Images for Multimodal Inputs and Outputs (Koh et al., 2023)

Inference


<img>
How
should
I
...
LLM
I
think
...
[IMG1]
..
[IMG{r}]
Linear
Visual Encoder
Tokenizer
Decision Model
ret
gen
Linear
GILLMapper
Similarity
SD Image Decoder
How should I display these at the farmer's market?
Image #1
Caption #1
Image-Text Inputs
Image Candidates
Retrieved Image
Generated Image
Final Model Outputs:
I think they look best when they are on a tray with a little bit of space between them.

Evaluation: Contextual Image Generation

- Given a Visual Story, generate a relevant image
- Need to condition on long, temporally dependent text and image inputs

Visual Storytelling ([Huang et al., 2016](#))

Evaluation: Contextual Image Generation

- Given a Visual Story, generate a relevant image
- Need to condition on long, temporally dependent text and image inputs

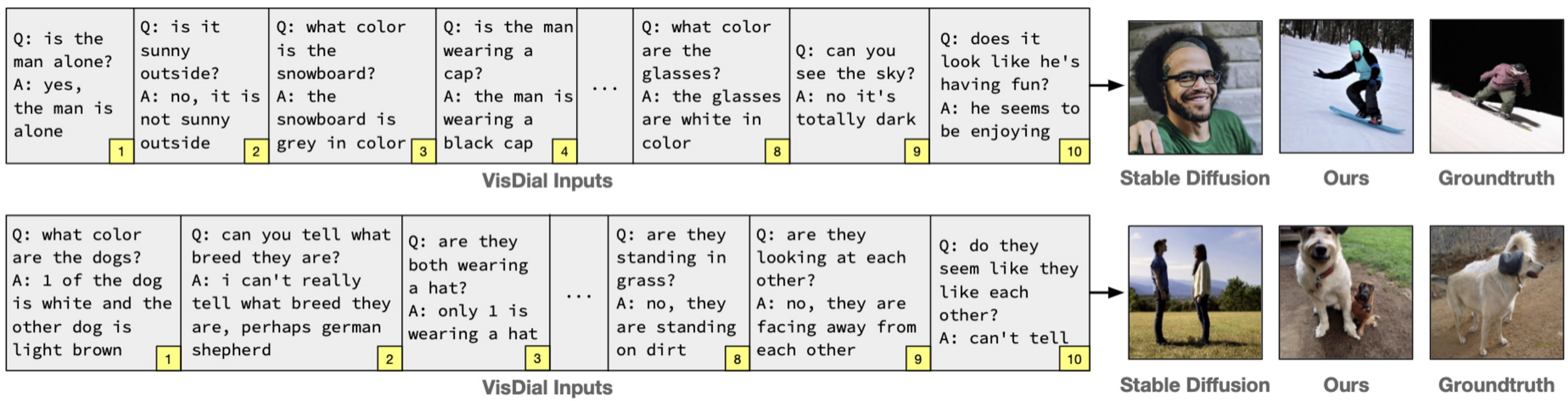
Model	CLIP Similarity (↑) 1 caption	CLIP Similarity (↑) 5 captions	CLIP Similarity (↑) 5 caps, 4 images	LPIPS (↓) 1 caption	LPIPS (↓) 5 captions	LPIPS (↓) 5 caps, 4 images
GLIDE [34]	0.582	0.591	-	0.753	0.745	-
Stable Diffusion [43]	**0.592** ±0.0007	0.598 ± 0.0006	-	0.703 ±0.0003	0.704 ± 0.0004	-
GILL	0.581 ±0.0005	**0.612** ±0.0011	**0.641** ±0.0011	**0.702** ±0.0004	**0.696** ±0.0008	**0.693** ±0.0008

- Model outperforms Stable Diffusion on longer input contexts
- GILL benefits from the abilities of the LLM (sensitivity to longer inputs, word orderings, in-context learning)

Visual Storytelling (Huang et al., 2016)

Evaluation: Contextual Image Generation

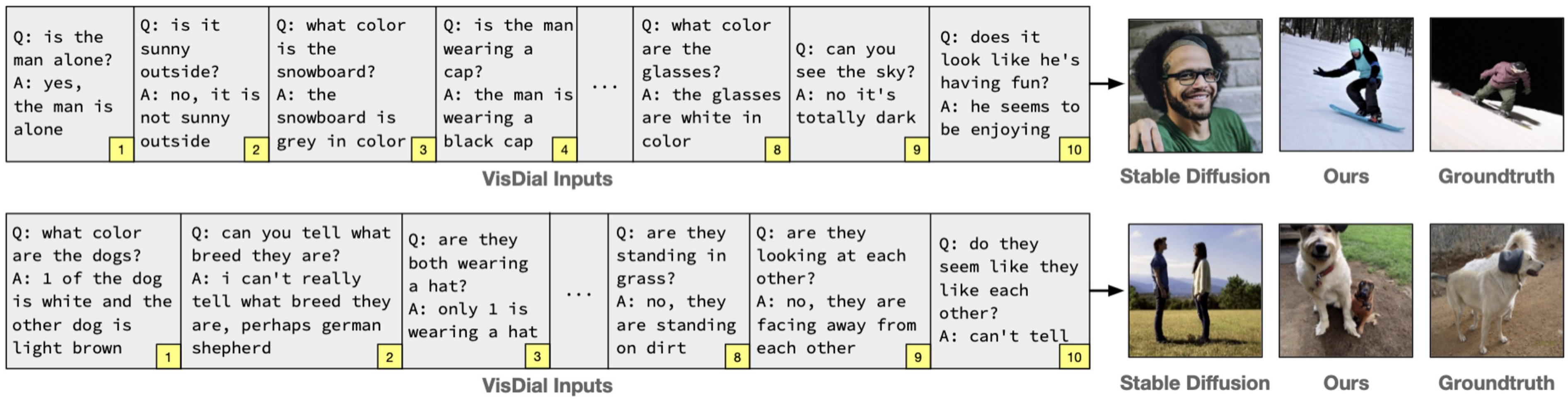
- Given a Visual Dialogue, generate a relevant image
- Need to condition on long dialogue-like text



Visual Dialog (Das et al., 2017)

Evaluation: Contextual Image Generation

- Given a Visual Dialogue, generate a relevant image
- Need to condition on long dialogue-like text



	CLIP Similarity (↑)			LPIPS (↓)		
Model	1 round	5 rounds	10 rounds	1 round	5 rounds	10 rounds
GLIDE [34]	**0.562**	0.595	0.587	0.800	0.794	0.799
Stable Diffusion [43]	0.552 ±0.0015	**0.629** ±0.0015	0.622 ±0.0012	**0.742** ±0.0010	0.722 ±0.0012	0.723 ±0.0008
GILL	0.528 ±0.0014	0.621 ±0.0009	**0.645** ±0.0010	**0.742** ±0.0022	**0.718** ±0.0028	**0.714** ±0.0006

The Effect of Context

Multi-modal context is **worth more** than uni-modal context, producing more relevant generation results.

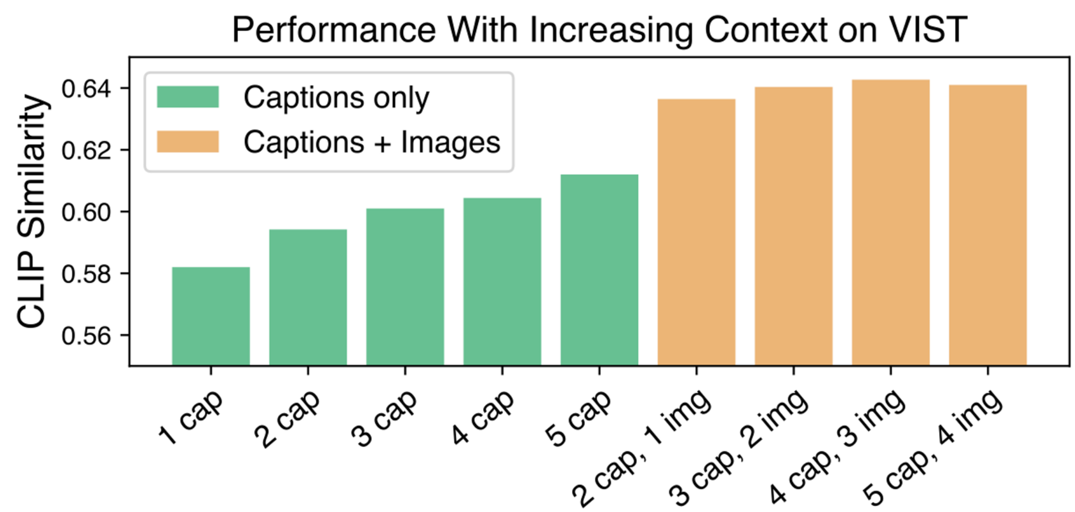
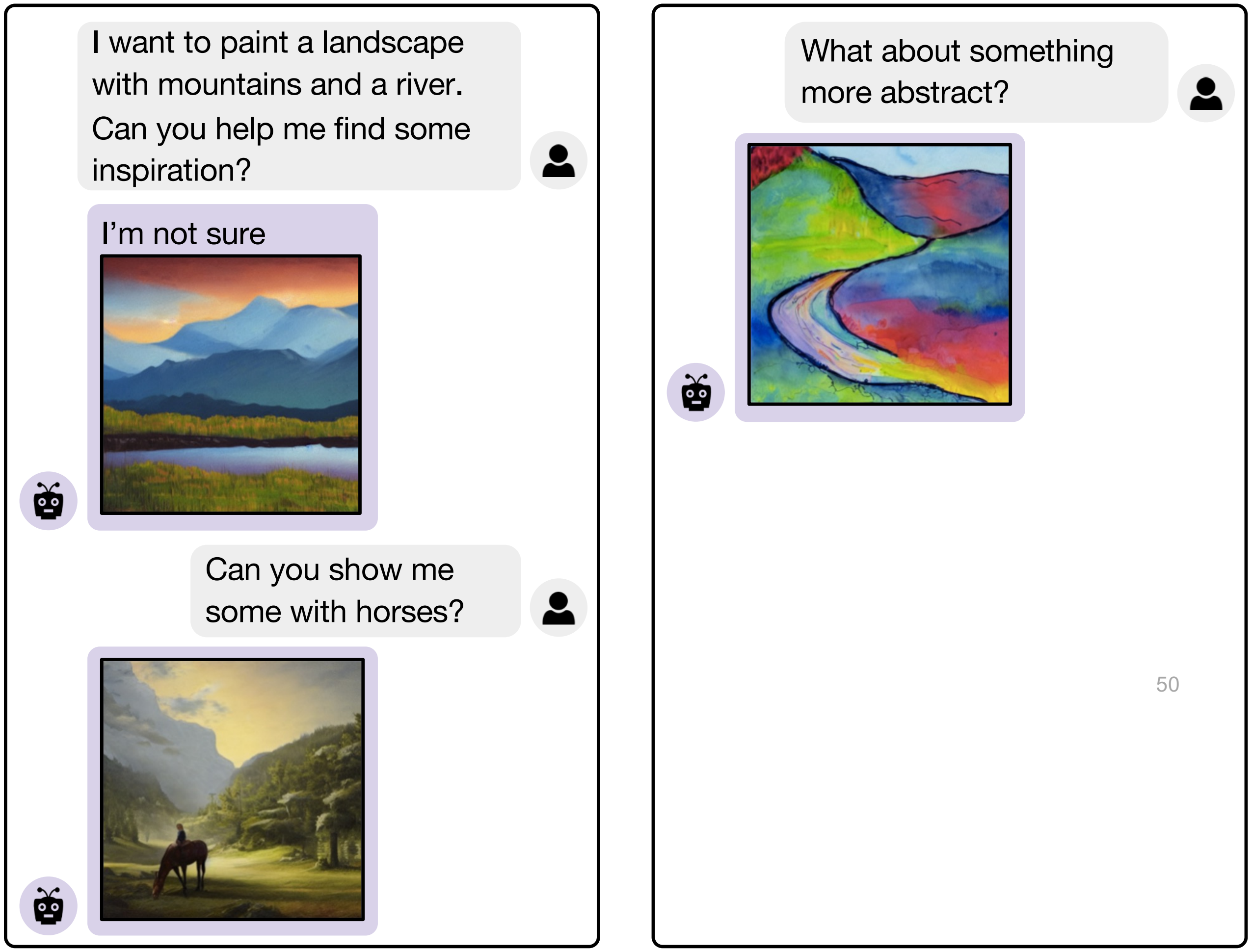


Image Refinement



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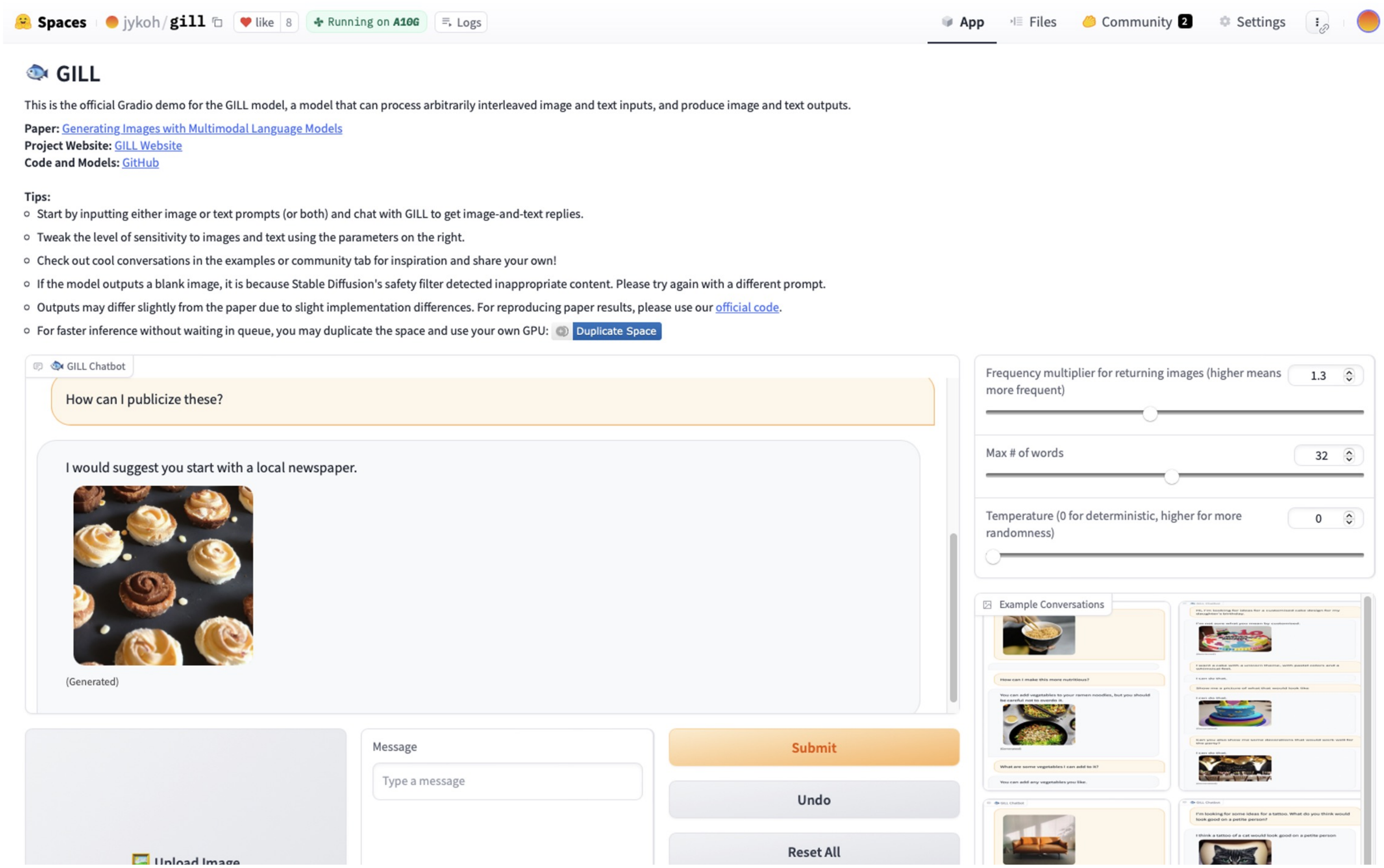
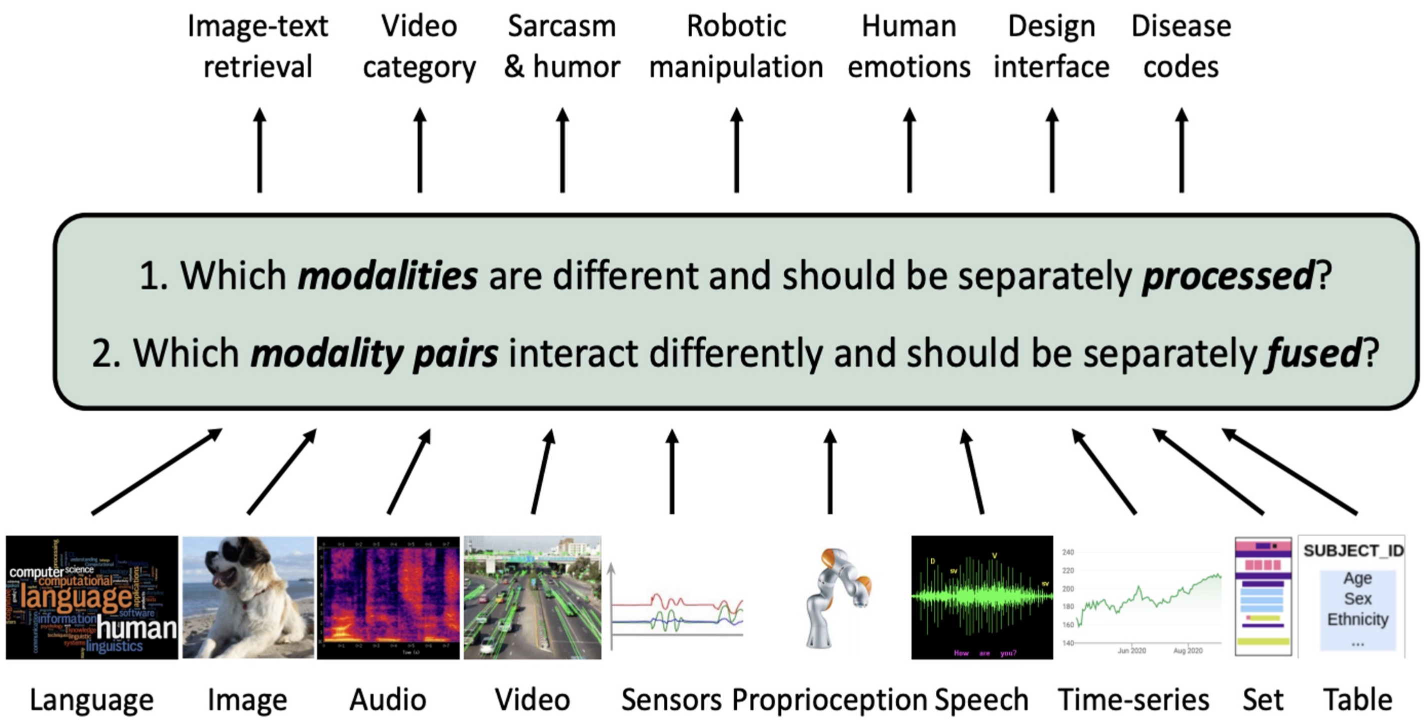
Try the Model

huggingface.co/spaces/jykoh/gill



Future Research: Incorporating More Modalities



High-Modality Multimodal Transformer: Quantifying Modality & Interaction Heterogeneity for High-Modality Representation Learning (Liang et al., 2023)

Future Research: Incorporating More Modalities

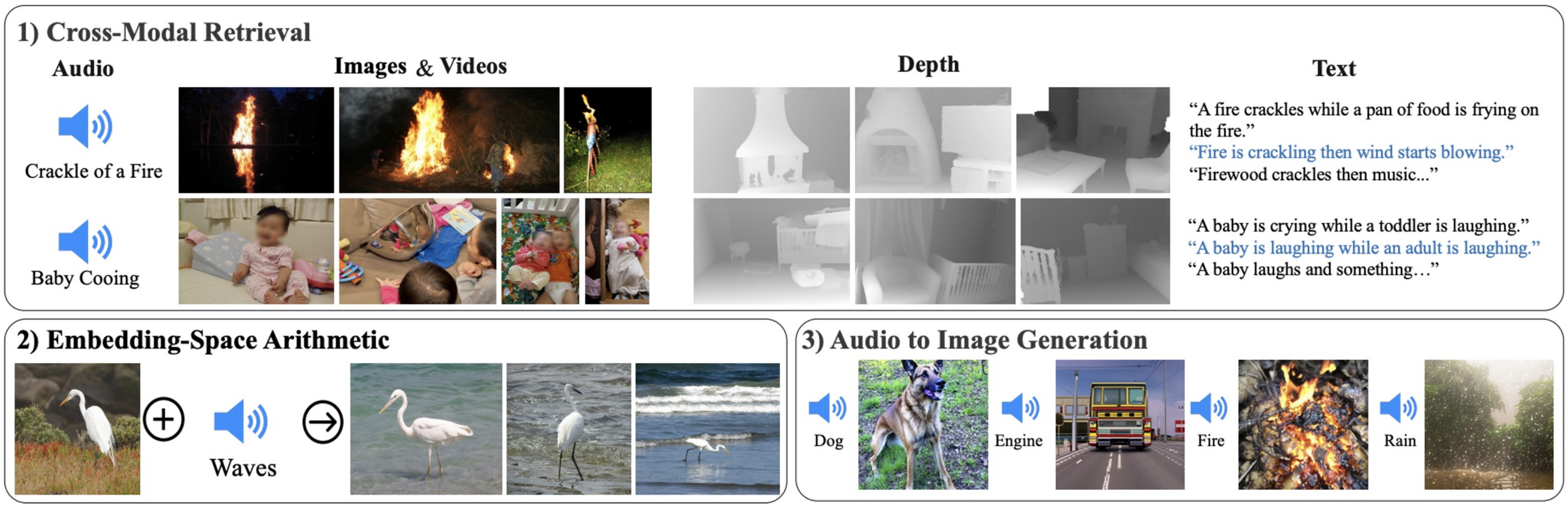
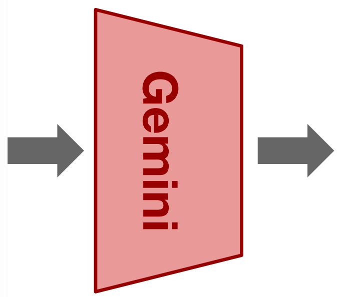


Figure 1. IMAGEBIND's joint embedding space enables novel multimodal capabilities. By aligning six modalities' embedding into a common space, IMAGEBIND enables: **1)** Cross-Modal Retrieval, which shows *emergent* alignment of modalities such as audio, depth or text, that aren't observed together. **2)** Adding embeddings from different modalities naturally composes their semantics. And **3)** Audio-to-Image generation, by using our audio embeddings with a pre-trained DALLE-2 [61] decoder designed to work with CLIP text embeddings.

IMAGEBIND: One Embedding Space To Bind Them All (Girdhar et al., 2023)

Future Research: Long Context Multimodal Models

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54

[The killer app of Gemini Pro 1.5 is video]

Future Research: Analysis on Architecture Design

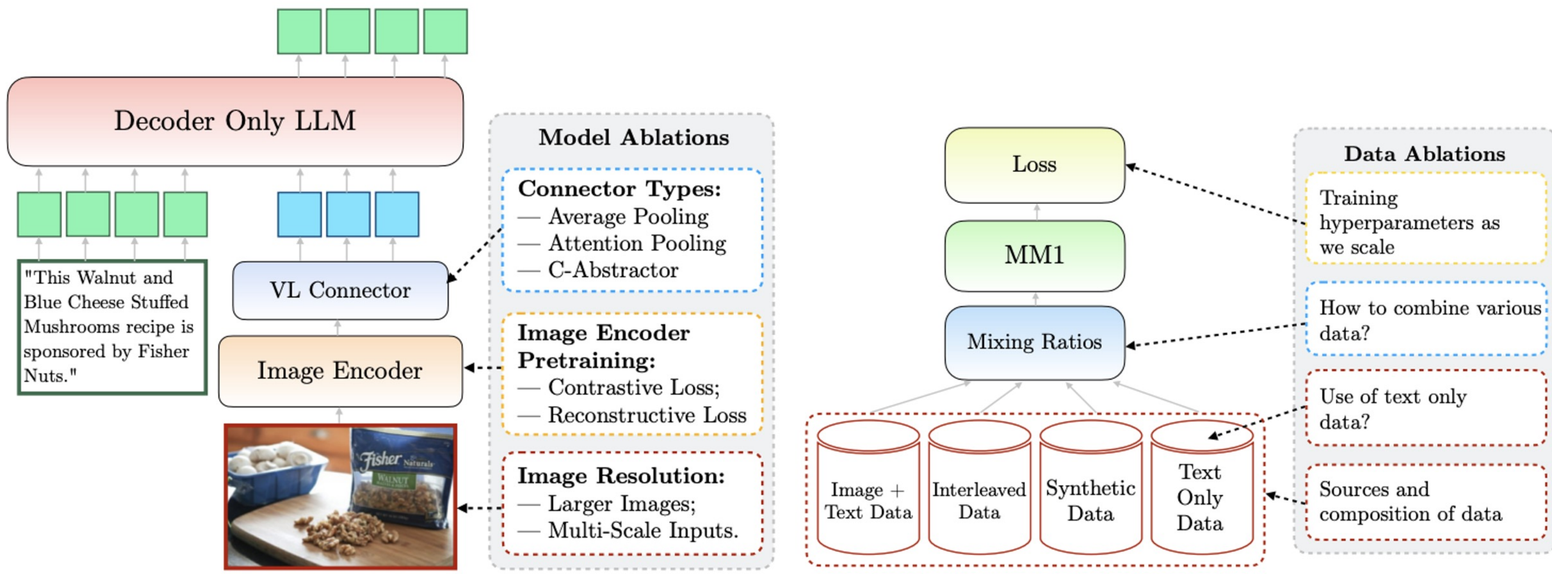
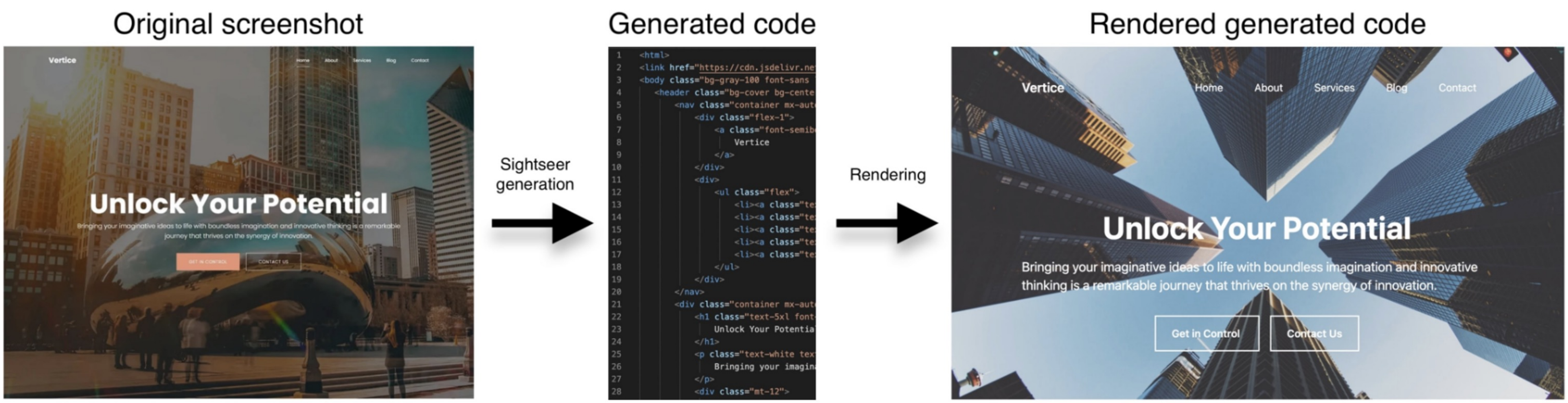


Fig. 3: *Left:* Model ablations: what visual encoder to use, how to feed rich visual data, and how to connect the visual representation to the LLM. *Right:* Data ablations: type of data, and their mixture.

MM1: Methods, Analysis & Insights from Multimodal LLM Pre-training (McKinzie et al., 2024)

Future Research: Multimodal Code Models



Unlocking the Conversion of Web Screenshots into HTML Code with the WebSight Dataset (Laurençon et al., 2024)
Design2Code: How Far Are We From Automating Front-End Engineering? (Si et al., 2024)

Other Resources

- **Blog post:** Multimodality and Large Multimodal Models (LMMs

- **Courses:** CMU 11-777: Multimodal Machine Learning
  - Lectures available on YouTube
  - 11-877: Advanced Topics in Multimodal Machine Learning

- **Survey papers:**
  - Foundations and Trends in Multimodal Machine Learning: Principles, Challenges, and Open Questions
  - A Survey on Multimodal Large Language Models
  - Multimodal Large Language Models: A Survey